

# Supplementary Material: NetShield

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## I. EVIDENCE CLASSIFICATION AND SCOPE

The primary Raspberry Pi campaign contains 12 sessions and 216 accepted arms. A separately frozen cyclic rotating full-admission (CRFA) extension contains 36 accepted arms over the same 12 session/workload identities and three capacities. The primary schedule-anchor reconstruction is derived from accepted primary hardware timestamps; the CRFA timing reconstruction is derived independently from its 9,720 accepted hardware occurrences. Queue-weight replay, phase-randomized control, rotating-frontier software conformance, and multi- $n$  scaling are read-only offline analyses and do not modify either hardware evidence chain.

TABLE I  
EVIDENCE CLASSES.

| Component                  | Evidence type                     | Permitted interpretation               |
|----------------------------|-----------------------------------|----------------------------------------|
| Primary campaign           | Hardware                          | Three-Pi A–F measurements              |
| CRFA extension             | Hardware                          | Three-Pi rotating-frontier measurement |
| Primary full-raw audit     | Raw reconstruction                | Accepted-arm traceability              |
| Queue-weight sweep         | Offline replay                    | Frozen-selector sensitivity            |
| Phase randomization        | Software sensitivity              | Desynchronization counterfactual       |
| Primary schedule anchors   | Hardware timestamp reconstruction | Post vs. nominal timing                |
| CRFA timing reconstruction | Hardware timestamp reconstruction | Frontier attenuation/timing            |
| Probe-count study          | Discrete-event conformance        | Analytical boundary check              |
| Rotating construction      | Discrete-event conformance        | Ideal byte-frontier construction       |

## II. FROZEN ACTION TABLE AND I-QPG THRESHOLDS

TABLE II  
FROZEN ACTION LATTICE AND  $v = 1$  THRESHOLDS AGAINST H-NONE.

| Action    | Payload (B) | Utility  | Threshold $Q$ (B) |
|-----------|-------------|----------|-------------------|
| L-NONE    | 0           | 0.758501 | –                 |
| L-ALERT   | 256         | 0.825168 | 221.84            |
| L-SUMMARY | 4096        | 0.891835 | 30.14             |
| M-NONE    | 0           | 0.766968 | –                 |
| M-ALERT   | 256         | 0.833635 | 254.91            |
| M-SUMMARY | 8192        | 0.900302 | 16.10             |
| H-NONE    | 0           | 0.768378 | –                 |
| H-ALERT   | 256         | 0.835044 | 260.41            |
| H-SUMMARY | 16384       | 0.901711 | 8.14              |
| H-FULL    | 131072      | 0.968378 | 1.53              |

## III. THEOREM DERIVATIONS

For  $m$  remotely represented probes and  $k$  full-evidence probes, payload is  $mS + k(B - S)$ . Hence one-round feasibility is

$$kB + (m - k)S \leq C,$$

and, for  $C \geq mS$ ,

$$k^*(C, m) = \min \left\{ m, \left\lfloor \frac{C - mS}{B - S} \right\rfloor \right\}.$$

With  $\rho = C/(nB)$ ,  $\sigma = S/B$ ,  $r = m/n$ , and  $f = k/n$ , the continuous frontier is  $\sigma r + (1 - \sigma)f \leq \rho$ .

If non-selected probes may send zero remote payload, payload for  $k$  full requests is minimized by setting  $m = k$ , so the unrestricted full-evidence byte frontier is

$$K_{\text{full}}^{\max} = \min\{n, \lfloor C/B \rfloor\}.$$

Provided  $C \geq nS$ , requiring all probes to remain remotely represented gives

$$K_{\text{all}}^{\max} = \left[ \left\lfloor \frac{C - nS}{B - S} \right\rfloor \right]_0^n.$$

If  $C < nS$ , the all-summary baseline is itself infeasible; clipping the count to zero does not make universal coverage feasible. For the formal  $n = 3$ ,  $B = 128$  KiB,  $S = 16$  KiB system, unrestricted versus all-probe counts are  $(1, 0)$ ,  $(2, 1)$ , and  $(3, 3)$  at low, mid, and high capacity. Thus universal per-round coverage costs one full-evidence slot at low and mid capacity.

For total admitted bytes  $A_t$  released at the start of round  $t$ , an idealized preemptive fluid server with common relative deadline  $D$  rounds satisfies the demand condition

$$\sum_{\tau=s}^t A_\tau \leq C(t - s + D), \quad s \leq t.$$

$D = 1$  reduces to per-round feasibility. Dividing the  $s = 1$  case by  $T$  gives the long-run load condition

$$\limsup_{T \rightarrow \infty} \frac{1}{T} \sum_{t=1}^T A_t \leq C.$$

The latter is a load-admissibility condition, not a deterministic bounded-delay guarantee for arbitrary burst processes.

For synchronized fixed-full overload from empty queues with  $nB > C$ , work-conserving service removes  $\min\{Q_\Sigma(t) + nB, C\} = C$  per round, giving  $Q_\Sigma(T) = T(nB - C)$ . At mid capacity, the post-full per-probe queue is  $B/3 = 43,690.6$  B, or approximately 42.7 KiB. This exceeds every positive-payload threshold; the next H-NONE round drains the residual, producing the frozen I-QPG full/none alternation.

In the reduced H-FULL/H-NONE desynchronized regime, if aggregate backlog is rate-stable and the server is asymptotically saturated, flow conservation gives

$$p^* = \min(1, C/(nB)) = \min(1, \rho).$$

The phase-randomized software study matches  $1/3$ ,  $2/3$ , and  $1$  at the three capacities.

Finally, with  $K = K_{\text{full}}^{\max}$ , selecting probes  $\mathcal{S}_t = \{(tK + j) \bmod n : j = 0, \dots, K - 1\}$  attains the unrestricted byte frontier every round. Its period is  $n/\gcd(n, K)$ , and each probe is selected  $K/\gcd(n, K)$  times per period.

#### IV. ANALYTICAL EXTENSIONS FOR DEPLOYMENT

These extensions derive planning consequences of the one-round byte model. They are not new hardware measurements, fitted arrival models, or guarantees of physical deadline attainment. MB denotes  $10^6$  bytes; KiB and MiB denote  $2^{10}$  and  $2^{20}$  bytes. The integer boundaries remain the operational reference for the three-probe measurements.

##### A. Probabilistic Requests and Capacity Planning

Assume empty incoming queues, fixed  $0 < S < B$ , and a budget  $C \geq nS$  that can first reserve summaries for all  $n$  probes. Within a round, each probe independently requests a full upload with common probability  $p \in [0, 1]$ ; otherwise it requests a summary. Independence across rounds is not needed for the following marginal per-round calculation. Let

$$J \sim \text{Binomial}(n, p), \quad h(C) = \min \left\{ n, \left\lfloor \frac{C - nS}{B - S} \right\rfloor \right\}. \quad (1)$$

Under a count-maximizing summary-preserving admission rule, the admitted full count is  $J_{\text{adm}} = \min\{J, h(C)\}$ . Consequently,

$$P_{\text{deg}}(C) = \Pr\{J > h(C)\} = \sum_{j=h(C)+1}^n \binom{n}{j} p^j (1-p)^{n-j}, \quad (2)$$

$$\mathbb{E}[J_{\text{adm}}] = \sum_{j=0}^n \min\{j, h(C)\} \binom{n}{j} p^j (1-p)^{n-j}. \quad (3)$$

Here  $P_{\text{deg}}$  is the probability that a round needs at least one full-to-summary downgrade, not a per-request loss probability. The expected number of downgraded requests is  $np - \mathbb{E}[J_{\text{adm}}]$ . Non-requesting and downgraded probes still retain summary evidence. Both equations concern admitted bytes, not returned core results.

For a target  $\epsilon \in [0, 1]$ , define the upper-tail quantile

$$q_\epsilon = \min\{q \in \{0, \dots, n\} : \Pr(J > q) \leq \epsilon\}. \quad (4)$$

The minimum budget within this model that preserves all summaries and satisfies  $P_{\text{deg}} \leq \epsilon$  is

$$\boxed{C_\epsilon = nS + (B - S)q_\epsilon}. \quad (5)$$

Indeed,  $h(C_\epsilon) = q_\epsilon$ . Any smaller budget either fails the summary baseline (when  $q_\epsilon = 0$ ) or admits fewer than  $q_\epsilon$  full requests, contradicting quantile minimality. Since  $h(C)$  is non-decreasing, expected admitted richness is also non-decreasing in  $C$ .

a) *Illustrative calculation, not an empirical result:* With the paper's  $n = 3$ ,  $B = 131,072$  B and  $S = 16,384$  B, suppose independently  $p = 2/3$  and choose  $\epsilon = 1/3$ . Then  $\Pr(J > 1) = 20/27$  but  $\Pr(J > 2) = 8/27$ , giving  $q_\epsilon = 2$  and  $C_\epsilon = 278,528$  B (272 KiB). At this budget,  $\mathbb{E}[J_{\text{adm}}] = 46/27$  and the expected downgrade count is  $8/27$  per round. The planning budget differs from the hardware mid budget of 256 KiB because it reserves a summary for the third probe. The numerical choice  $p = 2/3$  does not establish independence of the QPG traces or identify this planning rule with CRFA, which assigns H-NONE to non-selected probes.

For a large- $n$  interpretation, hold  $B, S, p$  and normalized capacity  $\rho = C_n/(nB)$  fixed, with  $\rho \geq \sigma = S/B$ . Then

$$\frac{\mathbb{E}[J_{\text{adm}}]}{n} \rightarrow \min \left\{ p, \min \left( 1, \frac{\rho - \sigma}{1 - \sigma} \right) \right\}. \quad (6)$$

This follows because  $J/n \rightarrow p$  in probability,  $h(C_n)/n$  converges to the clipped capacity term, and the normalized admitted count is bounded by one. For  $\rho < \sigma$ , universal summary coverage is infeasible rather than a zero-richness feasible point. With correlated within-round requests, the binomial tail is not justified; the same quantile construction requires the actual joint request-count distribution. Queueing, security utility, and physical timing remain outside this planning calculation.

### B. Heterogeneous Payloads: Summary Baseline and Upgrade Costs

Fix the set  $\mathcal{M}$  of probes required to retain remote evidence, with  $0 < S_i < B_i$  for every  $i \in \mathcal{M}$ . Let

$$S_{\mathcal{M}} = \sum_{i \in \mathcal{M}} S_i, \quad d_i = B_i - S_i > 0, \quad R = C - S_{\mathcal{M}}. \quad (7)$$

If  $R < 0$ , the prescribed coverage set is infeasible. Otherwise, upgrading  $\mathcal{F} \subseteq \mathcal{M}$  to full evidence is feasible exactly when

$$S_{\mathcal{M}} + \sum_{i \in \mathcal{F}} d_i \leq C. \quad (8)$$

For a pure full-count objective (equal positive value per upgrade), sort  $d_{(1)} \leq \dots \leq d_{(m)}$ , where  $m = |\mathcal{M}|$ . The optimum is

$$k_{\text{het}}^* = \max \left\{ k \in \{0, \dots, m\} : \sum_{j=1}^k d_{(j)} \leq R \right\}. \quad (9)$$

Any set of  $k$  upgrades costs at least the sum of the  $k$  smallest costs, which proves feasibility and optimality of this count rule. When payloads are homogeneous, it reduces to  $\min\{m, \lfloor (C - mS)/(B - S) \rfloor\}$  for  $C \geq mS$ .

For general nonnegative values  $w_i$ , the formulation is instead

$$\max_{x_i \in \{0,1\}} \sum_{i \in \mathcal{M}} w_i x_i \quad \text{subject to} \quad \sum_{i \in \mathcal{M}} d_i x_i \leq R, \quad (10)$$

a 0/1 knapsack problem. Equal value is not equal value density. With  $R = 3$  and cost/value pairs  $(2, 2)$  and  $(3, 3)$ , both densities equal one, but choosing the cheaper upgrade yields value 2 while choosing the other yields value 3. Thus equal density does not justify an increasing-cost greedy rule for the weighted problem. No heterogeneous-payload hardware validation or security-value optimum is claimed here; the scope is a fixed coverage set with separable values and no additional fairness constraints.

### C. Continuous Exchange Rate versus Discrete Feasibility

In the continuous relaxation,  $0 \leq f \leq r \leq 1$  and  $\sigma r + (1 - \sigma)f \leq \rho$ . Along any non-degenerate feasible segment of the active budget line,

$$\frac{dr}{df} = -\frac{1 - \sigma}{\sigma}. \quad (11)$$

This is a local coverage–richness exchange rate, not an integer scheduling rule. With  $\sigma = 1/8$ , its magnitude is seven. The full-coverage boundary point  $r = 1$ ,  $f = (\rho - \sigma)/(1 - \sigma)$  applies only for  $\sigma \leq \rho \leq 1$ ; for  $\rho > 1$ ,  $(r, f) = (1, 1)$  is already maximal although the byte budget is slack.

Discrete Pareto points can likewise leave unused capacity. At  $n = 3$  and  $C = B$  with  $S = B/8$ , the all-summary allocation  $(m, k) = (3, 0)$  costs only  $3S = 3B/8$ . It nevertheless cannot increase  $k$  while retaining  $m = 3$ , since one full upload plus two summaries costs  $B + 2S > B$ . It is therefore undominated in the integer coverage/count ordering despite its slack byte budget. For the hardware testbed, the integer slot price and CRFA selection schedule, rather than a continuous Pareto assertion, govern the feasible operating points.

## V. COMPLETE QUEUE-WEIGHT SWEEP RESULTS

The  $v = 1$  identity gate covers 36 accepted physical D cells and four deterministic checks per cell. Low-capacity cycle/action composition changes with  $v$ ; no all- $v$  cycle-identity claim is made.

TABLE III: Complete frozen-selector  $v$ -sweep results (432 retained rows).

| Session | $v$ | Cap. | Core frac. | Payload (B) | Max Q (B) | Period |
|---------|-----|------|------------|-------------|-----------|--------|
| F12_S01 | 1   | low  | 0.244444   | 9191424     | 87382     | 4      |
| F12_S02 | 1   | low  | 0.244444   | 9191424     | 87382     | 4      |
| F12_S03 | 1   | low  | 0.244444   | 9191424     | 87382     | 4      |
| F12_S04 | 1   | low  | 0.244444   | 9191424     | 87382     | 4      |
| F12_S05 | 1   | low  | 0.244444   | 9191424     | 87382     | 4      |
| F12_S06 | 1   | low  | 0.244444   | 9191424     | 87382     | 4      |
| F12_S07 | 1   | low  | 0.244444   | 9191424     | 87382     | 4      |
| F12_S08 | 1   | low  | 0.244444   | 9191424     | 87382     | 4      |
| F12_S09 | 1   | low  | 0.244444   | 9191424     | 87382     | 4      |
| F12_S10 | 1   | low  | 0.244444   | 9191424     | 87382     | 4      |
| F12_S11 | 1   | low  | 0.244444   | 9191424     | 87382     | 4      |
| F12_S12 | 1   | low  | 0.244444   | 9191424     | 87382     | 4      |
| F12_S01 | 1   | mid  | 0.500000   | 17694720    | 43691     | 2      |
| F12_S02 | 1   | mid  | 0.500000   | 17694720    | 43691     | 2      |
| F12_S03 | 1   | mid  | 0.500000   | 17694720    | 43691     | 2      |
| F12_S04 | 1   | mid  | 0.500000   | 17694720    | 43691     | 2      |
| F12_S05 | 1   | mid  | 0.500000   | 17694720    | 43691     | 2      |
| F12_S06 | 1   | mid  | 0.500000   | 17694720    | 43691     | 2      |
| F12_S07 | 1   | mid  | 0.500000   | 17694720    | 43691     | 2      |
| F12_S08 | 1   | mid  | 0.500000   | 17694720    | 43691     | 2      |
| F12_S09 | 1   | mid  | 0.500000   | 17694720    | 43691     | 2      |
| F12_S10 | 1   | mid  | 0.500000   | 17694720    | 43691     | 2      |
| F12_S11 | 1   | mid  | 0.500000   | 17694720    | 43691     | 2      |
| F12_S12 | 1   | mid  | 0.500000   | 17694720    | 43691     | 2      |
| F12_S01 | 1   | high | 1.000000   | 35389440    | 0         | 1      |
| F12_S02 | 1   | high | 1.000000   | 35389440    | 0         | 1      |
| F12_S03 | 1   | high | 1.000000   | 35389440    | 0         | 1      |
| F12_S04 | 1   | high | 1.000000   | 35389440    | 0         | 1      |
| F12_S05 | 1   | high | 1.000000   | 35389440    | 0         | 1      |
| F12_S06 | 1   | high | 1.000000   | 35389440    | 0         | 1      |
| F12_S07 | 1   | high | 1.000000   | 35389440    | 0         | 1      |
| F12_S08 | 1   | high | 1.000000   | 35389440    | 0         | 1      |
| F12_S09 | 1   | high | 1.000000   | 35389440    | 0         | 1      |
| F12_S10 | 1   | high | 1.000000   | 35389440    | 0         | 1      |
| F12_S11 | 1   | high | 1.000000   | 35389440    | 0         | 1      |
| F12_S12 | 1   | high | 1.000000   | 35389440    | 0         | 1      |
| F12_S01 | 3   | low  | 0.244444   | 9191424     | 87382     | 4      |
| F12_S02 | 3   | low  | 0.244444   | 9191424     | 87382     | 4      |
| F12_S03 | 3   | low  | 0.244444   | 9191424     | 87382     | 4      |
| F12_S04 | 3   | low  | 0.244444   | 9191424     | 87382     | 4      |
| F12_S05 | 3   | low  | 0.244444   | 9191424     | 87382     | 4      |
| F12_S06 | 3   | low  | 0.244444   | 9191424     | 87382     | 4      |
| F12_S07 | 3   | low  | 0.244444   | 9191424     | 87382     | 4      |
| F12_S08 | 3   | low  | 0.244444   | 9191424     | 87382     | 4      |
| F12_S09 | 3   | low  | 0.244444   | 9191424     | 87382     | 4      |
| F12_S10 | 3   | low  | 0.244444   | 9191424     | 87382     | 4      |
| F12_S11 | 3   | low  | 0.244444   | 9191424     | 87382     | 4      |
| F12_S12 | 3   | low  | 0.244444   | 9191424     | 87382     | 4      |
| F12_S01 | 3   | mid  | 0.500000   | 17694720    | 43691     | 2      |
| F12_S02 | 3   | mid  | 0.500000   | 17694720    | 43691     | 2      |
| F12_S03 | 3   | mid  | 0.500000   | 17694720    | 43691     | 2      |
| F12_S04 | 3   | mid  | 0.500000   | 17694720    | 43691     | 2      |
| F12_S05 | 3   | mid  | 0.500000   | 17694720    | 43691     | 2      |
| F12_S06 | 3   | mid  | 0.500000   | 17694720    | 43691     | 2      |

| Session | $v$ | Cap. | Core frac. | Payload (B) | Max Q (B) | Period |
|---------|-----|------|------------|-------------|-----------|--------|
| F12_S07 | 3   | mid  | 0.500000   | 17694720    | 43691     | 2      |
| F12_S08 | 3   | mid  | 0.500000   | 17694720    | 43691     | 2      |
| F12_S09 | 3   | mid  | 0.500000   | 17694720    | 43691     | 2      |
| F12_S10 | 3   | mid  | 0.500000   | 17694720    | 43691     | 2      |
| F12_S11 | 3   | mid  | 0.500000   | 17694720    | 43691     | 2      |
| F12_S12 | 3   | mid  | 0.500000   | 17694720    | 43691     | 2      |
| F12_S01 | 3   | high | 1.000000   | 35389440    | 0         | 1      |
| F12_S02 | 3   | high | 1.000000   | 35389440    | 0         | 1      |
| F12_S03 | 3   | high | 1.000000   | 35389440    | 0         | 1      |
| F12_S04 | 3   | high | 1.000000   | 35389440    | 0         | 1      |
| F12_S05 | 3   | high | 1.000000   | 35389440    | 0         | 1      |
| F12_S06 | 3   | high | 1.000000   | 35389440    | 0         | 1      |
| F12_S07 | 3   | high | 1.000000   | 35389440    | 0         | 1      |
| F12_S08 | 3   | high | 1.000000   | 35389440    | 0         | 1      |
| F12_S09 | 3   | high | 1.000000   | 35389440    | 0         | 1      |
| F12_S10 | 3   | high | 1.000000   | 35389440    | 0         | 1      |
| F12_S11 | 3   | high | 1.000000   | 35389440    | 0         | 1      |
| F12_S12 | 3   | high | 1.000000   | 35389440    | 0         | 1      |
| F12_S01 | 10  | low  | 0.300000   | 10838016    | 87386     | 10     |
| F12_S02 | 10  | low  | 0.300000   | 10838016    | 87386     | 10     |
| F12_S03 | 10  | low  | 0.300000   | 10838016    | 87386     | 10     |
| F12_S04 | 10  | low  | 0.300000   | 10838016    | 87386     | 10     |
| F12_S05 | 10  | low  | 0.300000   | 10838016    | 87386     | 10     |
| F12_S06 | 10  | low  | 0.300000   | 10838016    | 87386     | 10     |
| F12_S07 | 10  | low  | 0.300000   | 10838016    | 87386     | 10     |
| F12_S08 | 10  | low  | 0.300000   | 10838016    | 87386     | 10     |
| F12_S09 | 10  | low  | 0.300000   | 10838016    | 87386     | 10     |
| F12_S10 | 10  | low  | 0.300000   | 10838016    | 87386     | 10     |
| F12_S11 | 10  | low  | 0.300000   | 10838016    | 87386     | 10     |
| F12_S12 | 10  | low  | 0.300000   | 10838016    | 87386     | 10     |
| F12_S01 | 10  | mid  | 0.500000   | 17694720    | 43691     | 2      |
| F12_S02 | 10  | mid  | 0.500000   | 17694720    | 43691     | 2      |
| F12_S03 | 10  | mid  | 0.500000   | 17694720    | 43691     | 2      |
| F12_S04 | 10  | mid  | 0.500000   | 17694720    | 43691     | 2      |
| F12_S05 | 10  | mid  | 0.500000   | 17694720    | 43691     | 2      |
| F12_S06 | 10  | mid  | 0.500000   | 17694720    | 43691     | 2      |
| F12_S07 | 10  | mid  | 0.500000   | 17694720    | 43691     | 2      |
| F12_S08 | 10  | mid  | 0.500000   | 17694720    | 43691     | 2      |
| F12_S09 | 10  | mid  | 0.500000   | 17694720    | 43691     | 2      |
| F12_S10 | 10  | mid  | 0.500000   | 17694720    | 43691     | 2      |
| F12_S11 | 10  | mid  | 0.500000   | 17694720    | 43691     | 2      |
| F12_S12 | 10  | mid  | 0.500000   | 17694720    | 43691     | 2      |
| F12_S01 | 10  | high | 1.000000   | 35389440    | 0         | 1      |
| F12_S02 | 10  | high | 1.000000   | 35389440    | 0         | 1      |
| F12_S03 | 10  | high | 1.000000   | 35389440    | 0         | 1      |
| F12_S04 | 10  | high | 1.000000   | 35389440    | 0         | 1      |
| F12_S05 | 10  | high | 1.000000   | 35389440    | 0         | 1      |
| F12_S06 | 10  | high | 1.000000   | 35389440    | 0         | 1      |
| F12_S07 | 10  | high | 1.000000   | 35389440    | 0         | 1      |
| F12_S08 | 10  | high | 1.000000   | 35389440    | 0         | 1      |
| F12_S09 | 10  | high | 1.000000   | 35389440    | 0         | 1      |
| F12_S10 | 10  | high | 1.000000   | 35389440    | 0         | 1      |
| F12_S11 | 10  | high | 1.000000   | 35389440    | 0         | 1      |
| F12_S12 | 10  | high | 1.000000   | 35389440    | 0         | 1      |
| F12_S01 | 30  | low  | 0.322222   | 11476992    | 87398     | 28     |
| F12_S02 | 30  | low  | 0.322222   | 11476992    | 87398     | 28     |
| F12_S03 | 30  | low  | 0.322222   | 11476992    | 87398     | 28     |
| F12_S04 | 30  | low  | 0.322222   | 11476992    | 87398     | 28     |
| F12_S05 | 30  | low  | 0.322222   | 11476992    | 87398     | 28     |
| F12_S06 | 30  | low  | 0.322222   | 11476992    | 87398     | 28     |
| F12_S07 | 30  | low  | 0.322222   | 11476992    | 87398     | 28     |
| F12_S08 | 30  | low  | 0.322222   | 11476992    | 87398     | 28     |

| Session | $v$ | Cap. | Core frac. | Payload (B) | Max Q (B) | Period |
|---------|-----|------|------------|-------------|-----------|--------|
| F12_S09 | 30  | low  | 0.322222   | 11476992    | 87398     | 28     |
| F12_S10 | 30  | low  | 0.322222   | 11476992    | 87398     | 28     |
| F12_S11 | 30  | low  | 0.322222   | 11476992    | 87398     | 28     |
| F12_S12 | 30  | low  | 0.322222   | 11476992    | 87398     | 28     |
| F12_S01 | 30  | mid  | 0.500000   | 17694720    | 43691     | 2      |
| F12_S02 | 30  | mid  | 0.500000   | 17694720    | 43691     | 2      |
| F12_S03 | 30  | mid  | 0.500000   | 17694720    | 43691     | 2      |
| F12_S04 | 30  | mid  | 0.500000   | 17694720    | 43691     | 2      |
| F12_S05 | 30  | mid  | 0.500000   | 17694720    | 43691     | 2      |
| F12_S06 | 30  | mid  | 0.500000   | 17694720    | 43691     | 2      |
| F12_S07 | 30  | mid  | 0.500000   | 17694720    | 43691     | 2      |
| F12_S08 | 30  | mid  | 0.500000   | 17694720    | 43691     | 2      |
| F12_S09 | 30  | mid  | 0.500000   | 17694720    | 43691     | 2      |
| F12_S10 | 30  | mid  | 0.500000   | 17694720    | 43691     | 2      |
| F12_S11 | 30  | mid  | 0.500000   | 17694720    | 43691     | 2      |
| F12_S12 | 30  | mid  | 0.500000   | 17694720    | 43691     | 2      |
| F12_S01 | 30  | high | 1.000000   | 35389440    | 0         | 1      |
| F12_S02 | 30  | high | 1.000000   | 35389440    | 0         | 1      |
| F12_S03 | 30  | high | 1.000000   | 35389440    | 0         | 1      |
| F12_S04 | 30  | high | 1.000000   | 35389440    | 0         | 1      |
| F12_S05 | 30  | high | 1.000000   | 35389440    | 0         | 1      |
| F12_S06 | 30  | high | 1.000000   | 35389440    | 0         | 1      |
| F12_S07 | 30  | high | 1.000000   | 35389440    | 0         | 1      |
| F12_S08 | 30  | high | 1.000000   | 35389440    | 0         | 1      |
| F12_S09 | 30  | high | 1.000000   | 35389440    | 0         | 1      |
| F12_S10 | 30  | high | 1.000000   | 35389440    | 0         | 1      |
| F12_S11 | 30  | high | 1.000000   | 35389440    | 0         | 1      |
| F12_S12 | 30  | high | 1.000000   | 35389440    | 0         | 1      |
| F12_S01 | 100 | low  | 0.333333   | 11821056    | 87436     | —      |
| F12_S02 | 100 | low  | 0.333333   | 11821056    | 87436     | —      |
| F12_S03 | 100 | low  | 0.333333   | 11821056    | 87436     | —      |
| F12_S04 | 100 | low  | 0.333333   | 11821056    | 87436     | —      |
| F12_S05 | 100 | low  | 0.333333   | 11821056    | 87436     | —      |
| F12_S06 | 100 | low  | 0.333333   | 11821056    | 87436     | —      |
| F12_S07 | 100 | low  | 0.333333   | 11821056    | 87436     | —      |
| F12_S08 | 100 | low  | 0.333333   | 11821056    | 87436     | —      |
| F12_S09 | 100 | low  | 0.333333   | 11821056    | 87436     | —      |
| F12_S10 | 100 | low  | 0.333333   | 11821056    | 87436     | —      |
| F12_S11 | 100 | low  | 0.333333   | 11821056    | 87436     | —      |
| F12_S12 | 100 | low  | 0.333333   | 11821056    | 87436     | —      |
| F12_S01 | 100 | mid  | 0.500000   | 17694720    | 43691     | 2      |
| F12_S02 | 100 | mid  | 0.500000   | 17694720    | 43691     | 2      |
| F12_S03 | 100 | mid  | 0.500000   | 17694720    | 43691     | 2      |
| F12_S04 | 100 | mid  | 0.500000   | 17694720    | 43691     | 2      |
| F12_S05 | 100 | mid  | 0.500000   | 17694720    | 43691     | 2      |
| F12_S06 | 100 | mid  | 0.500000   | 17694720    | 43691     | 2      |
| F12_S07 | 100 | mid  | 0.500000   | 17694720    | 43691     | 2      |
| F12_S08 | 100 | mid  | 0.500000   | 17694720    | 43691     | 2      |
| F12_S09 | 100 | mid  | 0.500000   | 17694720    | 43691     | 2      |
| F12_S10 | 100 | mid  | 0.500000   | 17694720    | 43691     | 2      |
| F12_S11 | 100 | mid  | 0.500000   | 17694720    | 43691     | 2      |
| F12_S12 | 100 | mid  | 0.500000   | 17694720    | 43691     | 2      |
| F12_S01 | 100 | high | 1.000000   | 35389440    | 0         | 1      |
| F12_S02 | 100 | high | 1.000000   | 35389440    | 0         | 1      |
| F12_S03 | 100 | high | 1.000000   | 35389440    | 0         | 1      |
| F12_S04 | 100 | high | 1.000000   | 35389440    | 0         | 1      |
| F12_S05 | 100 | high | 1.000000   | 35389440    | 0         | 1      |
| F12_S06 | 100 | high | 1.000000   | 35389440    | 0         | 1      |
| F12_S07 | 100 | high | 1.000000   | 35389440    | 0         | 1      |
| F12_S08 | 100 | high | 1.000000   | 35389440    | 0         | 1      |
| F12_S09 | 100 | high | 1.000000   | 35389440    | 0         | 1      |
| F12_S10 | 100 | high | 1.000000   | 35389440    | 0         | 1      |

| Session | $v$ | Cap. | Core frac. | Payload (B) | Max Q (B) | Period |
|---------|-----|------|------------|-------------|-----------|--------|
| F12_S11 | 100 | high | 1.000000   | 35389440    | 0         | 1      |
| F12_S12 | 100 | high | 1.000000   | 35389440    | 0         | 1      |
| F12_S01 | 168 | low  | 0.244444   | 9209088     | 87382     | 4      |
| F12_S02 | 168 | low  | 0.244444   | 9209088     | 87382     | 4      |
| F12_S03 | 168 | low  | 0.244444   | 9209088     | 87382     | 4      |
| F12_S04 | 168 | low  | 0.244444   | 9209088     | 87382     | 4      |
| F12_S05 | 168 | low  | 0.244444   | 9209088     | 87382     | 4      |
| F12_S06 | 168 | low  | 0.244444   | 9209088     | 87382     | 4      |
| F12_S07 | 168 | low  | 0.244444   | 9209088     | 87382     | 4      |
| F12_S08 | 168 | low  | 0.244444   | 9209088     | 87382     | 4      |
| F12_S09 | 168 | low  | 0.244444   | 9209088     | 87382     | 4      |
| F12_S10 | 168 | low  | 0.244444   | 9209088     | 87382     | 4      |
| F12_S11 | 168 | low  | 0.244444   | 9209088     | 87382     | 4      |
| F12_S12 | 168 | low  | 0.244444   | 9209088     | 87382     | 4      |
| F12_S01 | 168 | mid  | 0.500000   | 17729280    | 43691     | 2      |
| F12_S02 | 168 | mid  | 0.500000   | 17729280    | 43691     | 2      |
| F12_S03 | 168 | mid  | 0.500000   | 17729280    | 43691     | 2      |
| F12_S04 | 168 | mid  | 0.500000   | 17729280    | 43691     | 2      |
| F12_S05 | 168 | mid  | 0.500000   | 17729280    | 43691     | 2      |
| F12_S06 | 168 | mid  | 0.500000   | 17729280    | 43691     | 2      |
| F12_S07 | 168 | mid  | 0.500000   | 17729280    | 43691     | 2      |
| F12_S08 | 168 | mid  | 0.500000   | 17729280    | 43691     | 2      |
| F12_S09 | 168 | mid  | 0.500000   | 17729280    | 43691     | 2      |
| F12_S10 | 168 | mid  | 0.500000   | 17729280    | 43691     | 2      |
| F12_S11 | 168 | mid  | 0.500000   | 17729280    | 43691     | 2      |
| F12_S12 | 168 | mid  | 0.500000   | 17729280    | 43691     | 2      |
| F12_S01 | 168 | high | 1.000000   | 35389440    | 0         | 1      |
| F12_S02 | 168 | high | 1.000000   | 35389440    | 0         | 1      |
| F12_S03 | 168 | high | 1.000000   | 35389440    | 0         | 1      |
| F12_S04 | 168 | high | 1.000000   | 35389440    | 0         | 1      |
| F12_S05 | 168 | high | 1.000000   | 35389440    | 0         | 1      |
| F12_S06 | 168 | high | 1.000000   | 35389440    | 0         | 1      |
| F12_S07 | 168 | high | 1.000000   | 35389440    | 0         | 1      |
| F12_S08 | 168 | high | 1.000000   | 35389440    | 0         | 1      |
| F12_S09 | 168 | high | 1.000000   | 35389440    | 0         | 1      |
| F12_S10 | 168 | high | 1.000000   | 35389440    | 0         | 1      |
| F12_S11 | 168 | high | 1.000000   | 35389440    | 0         | 1      |
| F12_S12 | 168 | high | 1.000000   | 35389440    | 0         | 1      |
| F12_S01 | 300 | low  | 0.244444   | 9209088     | 87382     | 4      |
| F12_S02 | 300 | low  | 0.244444   | 9209088     | 87382     | 4      |
| F12_S03 | 300 | low  | 0.244444   | 9209088     | 87382     | 4      |
| F12_S04 | 300 | low  | 0.244444   | 9209088     | 87382     | 4      |
| F12_S05 | 300 | low  | 0.244444   | 9209088     | 87382     | 4      |
| F12_S06 | 300 | low  | 0.244444   | 9209088     | 87382     | 4      |
| F12_S07 | 300 | low  | 0.244444   | 9209088     | 87382     | 4      |
| F12_S08 | 300 | low  | 0.244444   | 9209088     | 87382     | 4      |
| F12_S09 | 300 | low  | 0.244444   | 9209088     | 87382     | 4      |
| F12_S10 | 300 | low  | 0.244444   | 9209088     | 87382     | 4      |
| F12_S11 | 300 | low  | 0.244444   | 9209088     | 87382     | 4      |
| F12_S12 | 300 | low  | 0.244444   | 9209088     | 87382     | 4      |
| F12_S01 | 300 | mid  | 0.500000   | 17729280    | 43691     | 2      |
| F12_S02 | 300 | mid  | 0.500000   | 17729280    | 43691     | 2      |
| F12_S03 | 300 | mid  | 0.500000   | 17729280    | 43691     | 2      |
| F12_S04 | 300 | mid  | 0.500000   | 17729280    | 43691     | 2      |
| F12_S05 | 300 | mid  | 0.500000   | 17729280    | 43691     | 2      |
| F12_S06 | 300 | mid  | 0.500000   | 17729280    | 43691     | 2      |
| F12_S07 | 300 | mid  | 0.500000   | 17729280    | 43691     | 2      |
| F12_S08 | 300 | mid  | 0.500000   | 17729280    | 43691     | 2      |
| F12_S09 | 300 | mid  | 0.500000   | 17729280    | 43691     | 2      |
| F12_S10 | 300 | mid  | 0.500000   | 17729280    | 43691     | 2      |
| F12_S11 | 300 | mid  | 0.500000   | 17729280    | 43691     | 2      |
| F12_S12 | 300 | mid  | 0.500000   | 17729280    | 43691     | 2      |

| Session | $v$  | Cap. | Core frac. | Payload (B) | Max Q (B) | Period |
|---------|------|------|------------|-------------|-----------|--------|
| F12_S01 | 300  | high | 1.000000   | 35389440    | 0         | 1      |
| F12_S02 | 300  | high | 1.000000   | 35389440    | 0         | 1      |
| F12_S03 | 300  | high | 1.000000   | 35389440    | 0         | 1      |
| F12_S04 | 300  | high | 1.000000   | 35389440    | 0         | 1      |
| F12_S05 | 300  | high | 1.000000   | 35389440    | 0         | 1      |
| F12_S06 | 300  | high | 1.000000   | 35389440    | 0         | 1      |
| F12_S07 | 300  | high | 1.000000   | 35389440    | 0         | 1      |
| F12_S08 | 300  | high | 1.000000   | 35389440    | 0         | 1      |
| F12_S09 | 300  | high | 1.000000   | 35389440    | 0         | 1      |
| F12_S10 | 300  | high | 1.000000   | 35389440    | 0         | 1      |
| F12_S11 | 300  | high | 1.000000   | 35389440    | 0         | 1      |
| F12_S12 | 300  | high | 1.000000   | 35389440    | 0         | 1      |
| F12_S01 | 1000 | low  | 0.288889   | 10582272    | 87896     | 7      |
| F12_S02 | 1000 | low  | 0.288889   | 10582272    | 87896     | 7      |
| F12_S03 | 1000 | low  | 0.288889   | 10582272    | 87896     | 7      |
| F12_S04 | 1000 | low  | 0.288889   | 10582272    | 87896     | 7      |
| F12_S05 | 1000 | low  | 0.288889   | 10582272    | 87896     | 7      |
| F12_S06 | 1000 | low  | 0.288889   | 10582272    | 87896     | 7      |
| F12_S07 | 1000 | low  | 0.288889   | 10582272    | 87896     | 7      |
| F12_S08 | 1000 | low  | 0.288889   | 10582272    | 87896     | 7      |
| F12_S09 | 1000 | low  | 0.288889   | 10582272    | 87896     | 7      |
| F12_S10 | 1000 | low  | 0.288889   | 10582272    | 87896     | 7      |
| F12_S11 | 1000 | low  | 0.288889   | 10582272    | 87896     | 7      |
| F12_S12 | 1000 | low  | 0.288889   | 10582272    | 87896     | 7      |
| F12_S01 | 1000 | mid  | 0.500000   | 17729280    | 43691     | 2      |
| F12_S02 | 1000 | mid  | 0.500000   | 17729280    | 43691     | 2      |
| F12_S03 | 1000 | mid  | 0.500000   | 17729280    | 43691     | 2      |
| F12_S04 | 1000 | mid  | 0.500000   | 17729280    | 43691     | 2      |
| F12_S05 | 1000 | mid  | 0.500000   | 17729280    | 43691     | 2      |
| F12_S06 | 1000 | mid  | 0.500000   | 17729280    | 43691     | 2      |
| F12_S07 | 1000 | mid  | 0.500000   | 17729280    | 43691     | 2      |
| F12_S08 | 1000 | mid  | 0.500000   | 17729280    | 43691     | 2      |
| F12_S09 | 1000 | mid  | 0.500000   | 17729280    | 43691     | 2      |
| F12_S10 | 1000 | mid  | 0.500000   | 17729280    | 43691     | 2      |
| F12_S11 | 1000 | mid  | 0.500000   | 17729280    | 43691     | 2      |
| F12_S12 | 1000 | mid  | 0.500000   | 17729280    | 43691     | 2      |
| F12_S01 | 1000 | high | 1.000000   | 35389440    | 0         | 1      |
| F12_S02 | 1000 | high | 1.000000   | 35389440    | 0         | 1      |
| F12_S03 | 1000 | high | 1.000000   | 35389440    | 0         | 1      |
| F12_S04 | 1000 | high | 1.000000   | 35389440    | 0         | 1      |
| F12_S05 | 1000 | high | 1.000000   | 35389440    | 0         | 1      |
| F12_S06 | 1000 | high | 1.000000   | 35389440    | 0         | 1      |
| F12_S07 | 1000 | high | 1.000000   | 35389440    | 0         | 1      |
| F12_S08 | 1000 | high | 1.000000   | 35389440    | 0         | 1      |
| F12_S09 | 1000 | high | 1.000000   | 35389440    | 0         | 1      |
| F12_S10 | 1000 | high | 1.000000   | 35389440    | 0         | 1      |
| F12_S11 | 1000 | high | 1.000000   | 35389440    | 0         | 1      |
| F12_S12 | 1000 | high | 1.000000   | 35389440    | 0         | 1      |
| F12_S01 | 3000 | low  | 0.244444   | 9491712     | 87382     | 4      |
| F12_S02 | 3000 | low  | 0.244444   | 9491712     | 87382     | 4      |
| F12_S03 | 3000 | low  | 0.244444   | 9491712     | 87382     | 4      |
| F12_S04 | 3000 | low  | 0.244444   | 9491712     | 87382     | 4      |
| F12_S05 | 3000 | low  | 0.244444   | 9491712     | 87382     | 4      |
| F12_S06 | 3000 | low  | 0.244444   | 9491712     | 87382     | 4      |
| F12_S07 | 3000 | low  | 0.244444   | 9491712     | 87382     | 4      |
| F12_S08 | 3000 | low  | 0.244444   | 9491712     | 87382     | 4      |
| F12_S09 | 3000 | low  | 0.244444   | 9491712     | 87382     | 4      |
| F12_S10 | 3000 | low  | 0.244444   | 9491712     | 87382     | 4      |
| F12_S11 | 3000 | low  | 0.244444   | 9491712     | 87382     | 4      |
| F12_S12 | 3000 | low  | 0.244444   | 9491712     | 87382     | 4      |
| F12_S01 | 3000 | mid  | 0.500000   | 18247680    | 43691     | 2      |
| F12_S02 | 3000 | mid  | 0.500000   | 18247680    | 43691     | 2      |

| Session | $v$   | Cap. | Core frac. | Payload (B) | Max Q (B) | Period |
|---------|-------|------|------------|-------------|-----------|--------|
| F12_S03 | 3000  | mid  | 0.500000   | 18247680    | 43691     | 2      |
| F12_S04 | 3000  | mid  | 0.500000   | 18247680    | 43691     | 2      |
| F12_S05 | 3000  | mid  | 0.500000   | 18247680    | 43691     | 2      |
| F12_S06 | 3000  | mid  | 0.500000   | 18247680    | 43691     | 2      |
| F12_S07 | 3000  | mid  | 0.500000   | 18247680    | 43691     | 2      |
| F12_S08 | 3000  | mid  | 0.500000   | 18247680    | 43691     | 2      |
| F12_S09 | 3000  | mid  | 0.500000   | 18247680    | 43691     | 2      |
| F12_S10 | 3000  | mid  | 0.500000   | 18247680    | 43691     | 2      |
| F12_S11 | 3000  | mid  | 0.500000   | 18247680    | 43691     | 2      |
| F12_S12 | 3000  | mid  | 0.500000   | 18247680    | 43691     | 2      |
| F12_S01 | 3000  | high | 1.000000   | 35389440    | 0         | 1      |
| F12_S02 | 3000  | high | 1.000000   | 35389440    | 0         | 1      |
| F12_S03 | 3000  | high | 1.000000   | 35389440    | 0         | 1      |
| F12_S04 | 3000  | high | 1.000000   | 35389440    | 0         | 1      |
| F12_S05 | 3000  | high | 1.000000   | 35389440    | 0         | 1      |
| F12_S06 | 3000  | high | 1.000000   | 35389440    | 0         | 1      |
| F12_S07 | 3000  | high | 1.000000   | 35389440    | 0         | 1      |
| F12_S08 | 3000  | high | 1.000000   | 35389440    | 0         | 1      |
| F12_S09 | 3000  | high | 1.000000   | 35389440    | 0         | 1      |
| F12_S10 | 3000  | high | 1.000000   | 35389440    | 0         | 1      |
| F12_S11 | 3000  | high | 1.000000   | 35389440    | 0         | 1      |
| F12_S12 | 3000  | high | 1.000000   | 35389440    | 0         | 1      |
| F12_S01 | 10000 | low  | 0.244444   | 9756672     | 87382     | 4      |
| F12_S02 | 10000 | low  | 0.244444   | 9756672     | 87382     | 4      |
| F12_S03 | 10000 | low  | 0.244444   | 9756672     | 87382     | 4      |
| F12_S04 | 10000 | low  | 0.244444   | 9756672     | 87382     | 4      |
| F12_S05 | 10000 | low  | 0.244444   | 9756672     | 87382     | 4      |
| F12_S06 | 10000 | low  | 0.244444   | 9756672     | 87382     | 4      |
| F12_S07 | 10000 | low  | 0.244444   | 9756672     | 87382     | 4      |
| F12_S08 | 10000 | low  | 0.244444   | 9756672     | 87382     | 4      |
| F12_S09 | 10000 | low  | 0.244444   | 9756672     | 87382     | 4      |
| F12_S10 | 10000 | low  | 0.244444   | 9756672     | 87382     | 4      |
| F12_S11 | 10000 | low  | 0.244444   | 9756672     | 87382     | 4      |
| F12_S12 | 10000 | low  | 0.244444   | 9756672     | 87382     | 4      |
| F12_S01 | 10000 | mid  | 0.500000   | 18247680    | 43691     | 2      |
| F12_S02 | 10000 | mid  | 0.500000   | 18247680    | 43691     | 2      |
| F12_S03 | 10000 | mid  | 0.500000   | 18247680    | 43691     | 2      |
| F12_S04 | 10000 | mid  | 0.500000   | 18247680    | 43691     | 2      |
| F12_S05 | 10000 | mid  | 0.500000   | 18247680    | 43691     | 2      |
| F12_S06 | 10000 | mid  | 0.500000   | 18247680    | 43691     | 2      |
| F12_S07 | 10000 | mid  | 0.500000   | 18247680    | 43691     | 2      |
| F12_S08 | 10000 | mid  | 0.500000   | 18247680    | 43691     | 2      |
| F12_S09 | 10000 | mid  | 0.500000   | 18247680    | 43691     | 2      |
| F12_S10 | 10000 | mid  | 0.500000   | 18247680    | 43691     | 2      |
| F12_S11 | 10000 | mid  | 0.500000   | 18247680    | 43691     | 2      |
| F12_S12 | 10000 | mid  | 0.500000   | 18247680    | 43691     | 2      |
| F12_S01 | 10000 | high | 1.000000   | 35389440    | 0         | 1      |
| F12_S02 | 10000 | high | 1.000000   | 35389440    | 0         | 1      |
| F12_S03 | 10000 | high | 1.000000   | 35389440    | 0         | 1      |
| F12_S04 | 10000 | high | 1.000000   | 35389440    | 0         | 1      |
| F12_S05 | 10000 | high | 1.000000   | 35389440    | 0         | 1      |
| F12_S06 | 10000 | high | 1.000000   | 35389440    | 0         | 1      |
| F12_S07 | 10000 | high | 1.000000   | 35389440    | 0         | 1      |
| F12_S08 | 10000 | high | 1.000000   | 35389440    | 0         | 1      |
| F12_S09 | 10000 | high | 1.000000   | 35389440    | 0         | 1      |
| F12_S10 | 10000 | high | 1.000000   | 35389440    | 0         | 1      |
| F12_S11 | 10000 | high | 1.000000   | 35389440    | 0         | 1      |
| F12_S12 | 10000 | high | 1.000000   | 35389440    | 0         | 1      |
| F12_S01 | 30000 | low  | 0.288889   | 11489280    | 99672     | 7      |
| F12_S02 | 30000 | low  | 0.288889   | 11489280    | 99672     | 7      |
| F12_S03 | 30000 | low  | 0.288889   | 11489280    | 99672     | 7      |
| F12_S04 | 30000 | low  | 0.288889   | 11489280    | 99672     | 7      |

| Session | $v$    | Cap. | Core frac. | Payload (B) | Max Q (B) | Period |
|---------|--------|------|------------|-------------|-----------|--------|
| F12_S05 | 30000  | low  | 0.288889   | 11489280    | 99672     | 7      |
| F12_S06 | 30000  | low  | 0.288889   | 11489280    | 99672     | 7      |
| F12_S07 | 30000  | low  | 0.288889   | 11489280    | 99672     | 7      |
| F12_S08 | 30000  | low  | 0.288889   | 11489280    | 99672     | 7      |
| F12_S09 | 30000  | low  | 0.288889   | 11489280    | 99672     | 7      |
| F12_S10 | 30000  | low  | 0.288889   | 11489280    | 99672     | 7      |
| F12_S11 | 30000  | low  | 0.288889   | 11489280    | 99672     | 7      |
| F12_S12 | 30000  | low  | 0.288889   | 11489280    | 99672     | 7      |
| F12_S01 | 30000  | mid  | 0.500000   | 18800640    | 43691     | 2      |
| F12_S02 | 30000  | mid  | 0.500000   | 18800640    | 43691     | 2      |
| F12_S03 | 30000  | mid  | 0.500000   | 18800640    | 43691     | 2      |
| F12_S04 | 30000  | mid  | 0.500000   | 18800640    | 43691     | 2      |
| F12_S05 | 30000  | mid  | 0.500000   | 18800640    | 43691     | 2      |
| F12_S06 | 30000  | mid  | 0.500000   | 18800640    | 43691     | 2      |
| F12_S07 | 30000  | mid  | 0.500000   | 18800640    | 43691     | 2      |
| F12_S08 | 30000  | mid  | 0.500000   | 18800640    | 43691     | 2      |
| F12_S09 | 30000  | mid  | 0.500000   | 18800640    | 43691     | 2      |
| F12_S10 | 30000  | mid  | 0.500000   | 18800640    | 43691     | 2      |
| F12_S11 | 30000  | mid  | 0.500000   | 18800640    | 43691     | 2      |
| F12_S12 | 30000  | mid  | 0.500000   | 18800640    | 43691     | 2      |
| F12_S01 | 30000  | high | 1.000000   | 35389440    | 0         | 1      |
| F12_S02 | 30000  | high | 1.000000   | 35389440    | 0         | 1      |
| F12_S03 | 30000  | high | 1.000000   | 35389440    | 0         | 1      |
| F12_S04 | 30000  | high | 1.000000   | 35389440    | 0         | 1      |
| F12_S05 | 30000  | high | 1.000000   | 35389440    | 0         | 1      |
| F12_S06 | 30000  | high | 1.000000   | 35389440    | 0         | 1      |
| F12_S07 | 30000  | high | 1.000000   | 35389440    | 0         | 1      |
| F12_S08 | 30000  | high | 1.000000   | 35389440    | 0         | 1      |
| F12_S09 | 30000  | high | 1.000000   | 35389440    | 0         | 1      |
| F12_S10 | 30000  | high | 1.000000   | 35389440    | 0         | 1      |
| F12_S11 | 30000  | high | 1.000000   | 35389440    | 0         | 1      |
| F12_S12 | 30000  | high | 1.000000   | 35389440    | 0         | 1      |
| F12_S01 | 100000 | low  | 0.288889   | 11796480    | 142052    | —      |
| F12_S02 | 100000 | low  | 0.288889   | 11796480    | 142052    | —      |
| F12_S03 | 100000 | low  | 0.288889   | 11796480    | 142052    | —      |
| F12_S04 | 100000 | low  | 0.288889   | 11796480    | 142052    | —      |
| F12_S05 | 100000 | low  | 0.288889   | 11796480    | 142052    | —      |
| F12_S06 | 100000 | low  | 0.288889   | 11796480    | 142052    | —      |
| F12_S07 | 100000 | low  | 0.288889   | 11796480    | 142052    | —      |
| F12_S08 | 100000 | low  | 0.288889   | 11796480    | 142052    | —      |
| F12_S09 | 100000 | low  | 0.288889   | 11796480    | 142052    | —      |
| F12_S10 | 100000 | low  | 0.288889   | 11796480    | 142052    | —      |
| F12_S11 | 100000 | low  | 0.288889   | 11796480    | 142052    | —      |
| F12_S12 | 100000 | low  | 0.288889   | 11796480    | 142052    | —      |
| F12_S01 | 100000 | mid  | 0.622222   | 22855680    | 95575     | 8      |
| F12_S02 | 100000 | mid  | 0.622222   | 22855680    | 95575     | 8      |
| F12_S03 | 100000 | mid  | 0.622222   | 22855680    | 95575     | 8      |
| F12_S04 | 100000 | mid  | 0.622222   | 22855680    | 95575     | 8      |
| F12_S05 | 100000 | mid  | 0.622222   | 22855680    | 95575     | 8      |
| F12_S06 | 100000 | mid  | 0.622222   | 22855680    | 95575     | 8      |
| F12_S07 | 100000 | mid  | 0.622222   | 22855680    | 95575     | 8      |
| F12_S08 | 100000 | mid  | 0.622222   | 22855680    | 95575     | 8      |
| F12_S09 | 100000 | mid  | 0.622222   | 22855680    | 95575     | 8      |
| F12_S10 | 100000 | mid  | 0.622222   | 22855680    | 95575     | 8      |
| F12_S11 | 100000 | mid  | 0.622222   | 22855680    | 95575     | 8      |
| F12_S12 | 100000 | mid  | 0.622222   | 22855680    | 95575     | 8      |
| F12_S01 | 100000 | high | 1.000000   | 35389440    | 0         | 1      |
| F12_S02 | 100000 | high | 1.000000   | 35389440    | 0         | 1      |
| F12_S03 | 100000 | high | 1.000000   | 35389440    | 0         | 1      |
| F12_S04 | 100000 | high | 1.000000   | 35389440    | 0         | 1      |
| F12_S05 | 100000 | high | 1.000000   | 35389440    | 0         | 1      |
| F12_S06 | 100000 | high | 1.000000   | 35389440    | 0         | 1      |

| Session | $v$    | Cap. | Core frac. | Payload (B) | Max Q (B) | Period |
|---------|--------|------|------------|-------------|-----------|--------|
| F12_S07 | 100000 | high | 1.000000   | 35389440    | 0         | 1      |
| F12_S08 | 100000 | high | 1.000000   | 35389440    | 0         | 1      |
| F12_S09 | 100000 | high | 1.000000   | 35389440    | 0         | 1      |
| F12_S10 | 100000 | high | 1.000000   | 35389440    | 0         | 1      |
| F12_S11 | 100000 | high | 1.000000   | 35389440    | 0         | 1      |
| F12_S12 | 100000 | high | 1.000000   | 35389440    | 0         | 1      |

## VI. COMPLETE PHASE-RANDOMIZATION SENSITIVITY

Probe phases are independently assigned from fixed seeds. Probes exchange neither action nor queue state, and no joint admission is introduced.

TABLE IV: Complete desynchronization software-sensitivity results (240 retained rows).

| Seed       | Cap. | Method                     | Core frac. | On-time  | Max Q (B) | Phases (ms) |
|------------|------|----------------------------|------------|----------|-----------|-------------|
| 2026090901 | low  | desynchronized_independent | 0.333333   | 0.000000 | 131029    | 595/193/691 |
| 2026090901 | low  | synchronized_independent   | 0.244444   | 0.000000 | 131029    | 0/0/0       |
| 2026090901 | low  | central_coordinated        | 0.000000   | –        | 16341     | 0/0/0       |
| 2026090901 | low  | shielded_coordinated       | 0.000000   | –        | 16341     | 0/0/0       |
| 2026090901 | mid  | desynchronized_independent | 0.666667   | 0.505556 | 130985    | 595/193/691 |
| 2026090901 | mid  | synchronized_independent   | 0.500000   | 0.000000 | 130985    | 0/0/0       |
| 2026090901 | mid  | central_coordinated        | 0.333333   | 1.000000 | 130985    | 0/0/0       |
| 2026090901 | mid  | shielded_coordinated       | 0.333333   | 1.000000 | 130985    | 0/0/0       |
| 2026090901 | high | desynchronized_independent | 1.000000   | 1.000000 | 121111    | 595/193/691 |
| 2026090901 | high | synchronized_independent   | 1.000000   | 1.000000 | 130941    | 0/0/0       |
| 2026090901 | high | central_coordinated        | 1.000000   | 1.000000 | 130941    | 0/0/0       |
| 2026090901 | high | shielded_coordinated       | 1.000000   | 1.000000 | 130941    | 0/0/0       |
| 2026090902 | low  | desynchronized_independent | 0.333333   | 0.000000 | 131029    | 403/632/306 |
| 2026090902 | low  | synchronized_independent   | 0.244444   | 0.000000 | 131029    | 0/0/0       |
| 2026090902 | low  | central_coordinated        | 0.000000   | –        | 16341     | 0/0/0       |
| 2026090902 | low  | shielded_coordinated       | 0.000000   | –        | 16341     | 0/0/0       |
| 2026090902 | mid  | desynchronized_independent | 0.666667   | 0.500000 | 130985    | 403/632/306 |
| 2026090902 | mid  | synchronized_independent   | 0.500000   | 0.000000 | 130985    | 0/0/0       |
| 2026090902 | mid  | central_coordinated        | 0.333333   | 1.000000 | 130985    | 0/0/0       |
| 2026090902 | mid  | shielded_coordinated       | 0.333333   | 1.000000 | 130985    | 0/0/0       |
| 2026090902 | high | desynchronized_independent | 1.000000   | 1.000000 | 130876    | 403/632/306 |
| 2026090902 | high | synchronized_independent   | 1.000000   | 1.000000 | 130941    | 0/0/0       |
| 2026090902 | high | central_coordinated        | 1.000000   | 1.000000 | 130941    | 0/0/0       |
| 2026090902 | high | shielded_coordinated       | 1.000000   | 1.000000 | 130941    | 0/0/0       |
| 2026090903 | low  | desynchronized_independent | 0.333333   | 0.000000 | 131029    | 489/690/996 |
| 2026090903 | low  | synchronized_independent   | 0.244444   | 0.000000 | 131029    | 0/0/0       |
| 2026090903 | low  | central_coordinated        | 0.000000   | –        | 16341     | 0/0/0       |
| 2026090903 | low  | shielded_coordinated       | 0.000000   | –        | 16341     | 0/0/0       |
| 2026090903 | mid  | desynchronized_independent | 0.666667   | 0.500000 | 130985    | 489/690/996 |
| 2026090903 | mid  | synchronized_independent   | 0.500000   | 0.000000 | 130985    | 0/0/0       |
| 2026090903 | mid  | central_coordinated        | 0.333333   | 1.000000 | 130985    | 0/0/0       |
| 2026090903 | mid  | shielded_coordinated       | 0.333333   | 1.000000 | 130985    | 0/0/0       |
| 2026090903 | high | desynchronized_independent | 1.000000   | 1.000000 | 1180      | 489/690/996 |
| 2026090903 | high | synchronized_independent   | 1.000000   | 1.000000 | 130941    | 0/0/0       |
| 2026090903 | high | central_coordinated        | 1.000000   | 1.000000 | 130941    | 0/0/0       |
| 2026090903 | high | shielded_coordinated       | 1.000000   | 1.000000 | 130941    | 0/0/0       |
| 2026090904 | low  | desynchronized_independent | 0.333333   | 0.000000 | 131028    | 945/680/799 |
| 2026090904 | low  | synchronized_independent   | 0.244444   | 0.000000 | 131029    | 0/0/0       |
| 2026090904 | low  | central_coordinated        | 0.000000   | –        | 16341     | 0/0/0       |
| 2026090904 | low  | shielded_coordinated       | 0.000000   | –        | 16341     | 0/0/0       |
| 2026090904 | mid  | desynchronized_independent | 0.666667   | 0.500000 | 130985    | 945/680/799 |
| 2026090904 | mid  | synchronized_independent   | 0.500000   | 0.000000 | 130985    | 0/0/0       |
| 2026090904 | mid  | central_coordinated        | 0.333333   | 1.000000 | 130985    | 0/0/0       |
| 2026090904 | mid  | shielded_coordinated       | 0.333333   | 1.000000 | 130985    | 0/0/0       |
| 2026090904 | high | desynchronized_independent | 1.000000   | 1.000000 | 21234     | 945/680/799 |
| 2026090904 | high | synchronized_independent   | 1.000000   | 1.000000 | 130941    | 0/0/0       |
| 2026090904 | high | central_coordinated        | 1.000000   | 1.000000 | 130941    | 0/0/0       |
| 2026090904 | high | shielded_coordinated       | 1.000000   | 1.000000 | 130941    | 0/0/0       |
| 2026090905 | low  | desynchronized_independent | 0.333333   | 0.000000 | 131028    | 154/172/828 |
| 2026090905 | low  | synchronized_independent   | 0.244444   | 0.000000 | 131029    | 0/0/0       |
| 2026090905 | low  | central_coordinated        | 0.000000   | –        | 16341     | 0/0/0       |
| 2026090905 | low  | shielded_coordinated       | 0.000000   | –        | 16341     | 0/0/0       |
| 2026090905 | mid  | desynchronized_independent | 0.666667   | 0.505556 | 130985    | 154/172/828 |
| 2026090905 | mid  | synchronized_independent   | 0.500000   | 0.000000 | 130985    | 0/0/0       |
| 2026090905 | mid  | central_coordinated        | 0.333333   | 1.000000 | 130985    | 0/0/0       |
| 2026090905 | mid  | shielded_coordinated       | 0.333333   | 1.000000 | 130985    | 0/0/0       |

| Seed       | Cap. | Method                     | Core frac. | On-time  | Max Q (B) | Phases (ms) |
|------------|------|----------------------------|------------|----------|-----------|-------------|
| 2026090905 | high | desynchronized_independent | 1.000000   | 1.000000 | 130876    | 154/172/828 |
| 2026090905 | high | synchronized_independent   | 1.000000   | 1.000000 | 130941    | 0/0/0       |
| 2026090905 | high | central_coordinated        | 1.000000   | 1.000000 | 130941    | 0/0/0       |
| 2026090905 | high | shielded_coordinated       | 1.000000   | 1.000000 | 130941    | 0/0/0       |
| 2026090906 | low  | desynchronized_independent | 0.333333   | 0.000000 | 131029    | 60/626/317  |
| 2026090906 | low  | synchronized_independent   | 0.244444   | 0.000000 | 131029    | 0/0/0       |
| 2026090906 | low  | central_coordinated        | 0.000000   | –        | 16341     | 0/0/0       |
| 2026090906 | low  | shielded_coordinated       | 0.000000   | –        | 16341     | 0/0/0       |
| 2026090906 | mid  | desynchronized_independent | 0.666667   | 0.505556 | 130985    | 60/626/317  |
| 2026090906 | mid  | synchronized_independent   | 0.500000   | 0.000000 | 130985    | 0/0/0       |
| 2026090906 | mid  | central_coordinated        | 0.333333   | 1.000000 | 130985    | 0/0/0       |
| 2026090906 | mid  | shielded_coordinated       | 0.333333   | 1.000000 | 130985    | 0/0/0       |
| 2026090906 | high | desynchronized_independent | 1.000000   | 1.000000 | 130876    | 60/626/317  |
| 2026090906 | high | synchronized_independent   | 1.000000   | 1.000000 | 130941    | 0/0/0       |
| 2026090906 | high | central_coordinated        | 1.000000   | 1.000000 | 130941    | 0/0/0       |
| 2026090906 | high | shielded_coordinated       | 1.000000   | 1.000000 | 130941    | 0/0/0       |
| 2026090907 | low  | desynchronized_independent | 0.333333   | 0.000000 | 131029    | 753/955/576 |
| 2026090907 | low  | synchronized_independent   | 0.244444   | 0.000000 | 131029    | 0/0/0       |
| 2026090907 | low  | central_coordinated        | 0.000000   | –        | 16341     | 0/0/0       |
| 2026090907 | low  | shielded_coordinated       | 0.000000   | –        | 16341     | 0/0/0       |
| 2026090907 | mid  | desynchronized_independent | 0.666667   | 0.500000 | 130985    | 753/955/576 |
| 2026090907 | mid  | synchronized_independent   | 0.500000   | 0.000000 | 130985    | 0/0/0       |
| 2026090907 | mid  | central_coordinated        | 0.333333   | 1.000000 | 130985    | 0/0/0       |
| 2026090907 | mid  | shielded_coordinated       | 0.333333   | 1.000000 | 130985    | 0/0/0       |
| 2026090907 | high | desynchronized_independent | 1.000000   | 1.000000 | 17302     | 753/955/576 |
| 2026090907 | high | synchronized_independent   | 1.000000   | 1.000000 | 130941    | 0/0/0       |
| 2026090907 | high | central_coordinated        | 1.000000   | 1.000000 | 130941    | 0/0/0       |
| 2026090907 | high | shielded_coordinated       | 1.000000   | 1.000000 | 130941    | 0/0/0       |
| 2026090908 | low  | desynchronized_independent | 0.333333   | 0.000000 | 131029    | 884/483/389 |
| 2026090908 | low  | synchronized_independent   | 0.244444   | 0.000000 | 131029    | 0/0/0       |
| 2026090908 | low  | central_coordinated        | 0.000000   | –        | 16341     | 0/0/0       |
| 2026090908 | low  | shielded_coordinated       | 0.000000   | –        | 16341     | 0/0/0       |
| 2026090908 | mid  | desynchronized_independent | 0.666667   | 0.500000 | 130985    | 884/483/389 |
| 2026090908 | mid  | synchronized_independent   | 0.500000   | 0.000000 | 130985    | 0/0/0       |
| 2026090908 | mid  | central_coordinated        | 0.333333   | 1.000000 | 130985    | 0/0/0       |
| 2026090908 | mid  | shielded_coordinated       | 0.333333   | 1.000000 | 130985    | 0/0/0       |
| 2026090908 | high | desynchronized_independent | 1.000000   | 1.000000 | 71828     | 884/483/389 |
| 2026090908 | high | synchronized_independent   | 1.000000   | 1.000000 | 130941    | 0/0/0       |
| 2026090908 | high | central_coordinated        | 1.000000   | 1.000000 | 130941    | 0/0/0       |
| 2026090908 | high | shielded_coordinated       | 1.000000   | 1.000000 | 130941    | 0/0/0       |
| 2026090909 | low  | desynchronized_independent | 0.333333   | 0.000000 | 131029    | 530/348/834 |
| 2026090909 | low  | synchronized_independent   | 0.244444   | 0.000000 | 131029    | 0/0/0       |
| 2026090909 | low  | central_coordinated        | 0.000000   | –        | 16341     | 0/0/0       |
| 2026090909 | low  | shielded_coordinated       | 0.000000   | –        | 16341     | 0/0/0       |
| 2026090909 | mid  | desynchronized_independent | 0.666667   | 0.500000 | 130985    | 530/348/834 |
| 2026090909 | mid  | synchronized_independent   | 0.500000   | 0.000000 | 130985    | 0/0/0       |
| 2026090909 | mid  | central_coordinated        | 0.333333   | 1.000000 | 130985    | 0/0/0       |
| 2026090909 | mid  | shielded_coordinated       | 0.333333   | 1.000000 | 130985    | 0/0/0       |
| 2026090909 | high | desynchronized_independent | 1.000000   | 1.000000 | 64881     | 530/348/834 |
| 2026090909 | high | synchronized_independent   | 1.000000   | 1.000000 | 130941    | 0/0/0       |
| 2026090909 | high | central_coordinated        | 1.000000   | 1.000000 | 130941    | 0/0/0       |
| 2026090909 | high | shielded_coordinated       | 1.000000   | 1.000000 | 130941    | 0/0/0       |
| 2026090910 | low  | desynchronized_independent | 0.333333   | 0.000000 | 131028    | 611/131/96  |
| 2026090910 | low  | synchronized_independent   | 0.244444   | 0.000000 | 131029    | 0/0/0       |
| 2026090910 | low  | central_coordinated        | 0.000000   | –        | 16341     | 0/0/0       |
| 2026090910 | low  | shielded_coordinated       | 0.000000   | –        | 16341     | 0/0/0       |
| 2026090910 | mid  | desynchronized_independent | 0.666667   | 0.505556 | 130985    | 611/131/96  |
| 2026090910 | mid  | synchronized_independent   | 0.500000   | 0.000000 | 130985    | 0/0/0       |
| 2026090910 | mid  | central_coordinated        | 0.333333   | 1.000000 | 130985    | 0/0/0       |
| 2026090910 | mid  | shielded_coordinated       | 0.333333   | 1.000000 | 130985    | 0/0/0       |
| 2026090910 | high | desynchronized_independent | 1.000000   | 1.000000 | 130876    | 611/131/96  |
| 2026090910 | high | synchronized_independent   | 1.000000   | 1.000000 | 130941    | 0/0/0       |

| Seed       | Cap. | Method                     | Core frac. | On-time  | Max Q (B) | Phases (ms) |
|------------|------|----------------------------|------------|----------|-----------|-------------|
| 2026090910 | high | central_coordinated        | 1.000000   | 1.000000 | 130941    | 0/0/0       |
| 2026090910 | high | shielded_coordinated       | 1.000000   | 1.000000 | 130941    | 0/0/0       |
| 2026090911 | low  | desynchronized_independent | 0.333333   | 0.000000 | 131029    | 286/697/613 |
| 2026090911 | low  | synchronized_independent   | 0.244444   | 0.000000 | 131029    | 0/0/0       |
| 2026090911 | low  | central_coordinated        | 0.000000   | –        | 16341     | 0/0/0       |
| 2026090911 | low  | shielded_coordinated       | 0.000000   | –        | 16341     | 0/0/0       |
| 2026090911 | mid  | desynchronized_independent | 0.666667   | 0.505556 | 130985    | 286/697/613 |
| 2026090911 | mid  | synchronized_independent   | 0.500000   | 0.000000 | 130985    | 0/0/0       |
| 2026090911 | mid  | central_coordinated        | 0.333333   | 1.000000 | 130985    | 0/0/0       |
| 2026090911 | mid  | shielded_coordinated       | 0.333333   | 1.000000 | 130985    | 0/0/0       |
| 2026090911 | high | desynchronized_independent | 1.000000   | 1.000000 | 118752    | 286/697/613 |
| 2026090911 | high | synchronized_independent   | 1.000000   | 1.000000 | 130941    | 0/0/0       |
| 2026090911 | high | central_coordinated        | 1.000000   | 1.000000 | 130941    | 0/0/0       |
| 2026090911 | high | shielded_coordinated       | 1.000000   | 1.000000 | 130941    | 0/0/0       |
| 2026090912 | low  | desynchronized_independent | 0.333333   | 0.000000 | 131028    | 72/898/820  |
| 2026090912 | low  | synchronized_independent   | 0.244444   | 0.000000 | 131029    | 0/0/0       |
| 2026090912 | low  | central_coordinated        | 0.000000   | –        | 16341     | 0/0/0       |
| 2026090912 | low  | shielded_coordinated       | 0.000000   | –        | 16341     | 0/0/0       |
| 2026090912 | mid  | desynchronized_independent | 0.666667   | 0.505556 | 130985    | 72/898/820  |
| 2026090912 | mid  | synchronized_independent   | 0.500000   | 0.000000 | 130985    | 0/0/0       |
| 2026090912 | mid  | central_coordinated        | 0.333333   | 1.000000 | 130985    | 0/0/0       |
| 2026090912 | mid  | shielded_coordinated       | 0.333333   | 1.000000 | 130985    | 0/0/0       |
| 2026090912 | high | desynchronized_independent | 1.000000   | 1.000000 | 102368    | 72/898/820  |
| 2026090912 | high | synchronized_independent   | 1.000000   | 1.000000 | 130941    | 0/0/0       |
| 2026090912 | high | central_coordinated        | 1.000000   | 1.000000 | 130941    | 0/0/0       |
| 2026090912 | high | shielded_coordinated       | 1.000000   | 1.000000 | 130941    | 0/0/0       |
| 2026090913 | low  | desynchronized_independent | 0.333333   | 0.000000 | 131028    | 31/997/848  |
| 2026090913 | low  | synchronized_independent   | 0.244444   | 0.000000 | 131029    | 0/0/0       |
| 2026090913 | low  | central_coordinated        | 0.000000   | –        | 16341     | 0/0/0       |
| 2026090913 | low  | shielded_coordinated       | 0.000000   | –        | 16341     | 0/0/0       |
| 2026090913 | mid  | desynchronized_independent | 0.666667   | 0.505556 | 130985    | 31/997/848  |
| 2026090913 | mid  | synchronized_independent   | 0.500000   | 0.000000 | 130985    | 0/0/0       |
| 2026090913 | mid  | central_coordinated        | 0.333333   | 1.000000 | 130985    | 0/0/0       |
| 2026090913 | mid  | shielded_coordinated       | 0.333333   | 1.000000 | 130985    | 0/0/0       |
| 2026090913 | high | desynchronized_independent | 1.000000   | 1.000000 | 118490    | 31/997/848  |
| 2026090913 | high | synchronized_independent   | 1.000000   | 1.000000 | 130941    | 0/0/0       |
| 2026090913 | high | central_coordinated        | 1.000000   | 1.000000 | 130941    | 0/0/0       |
| 2026090913 | high | shielded_coordinated       | 1.000000   | 1.000000 | 130941    | 0/0/0       |
| 2026090914 | low  | desynchronized_independent | 0.333333   | 0.000000 | 131029    | 379/396/678 |
| 2026090914 | low  | synchronized_independent   | 0.244444   | 0.000000 | 131029    | 0/0/0       |
| 2026090914 | low  | central_coordinated        | 0.000000   | –        | 16341     | 0/0/0       |
| 2026090914 | low  | shielded_coordinated       | 0.000000   | –        | 16341     | 0/0/0       |
| 2026090914 | mid  | desynchronized_independent | 0.666667   | 0.500000 | 130985    | 379/396/678 |
| 2026090914 | mid  | synchronized_independent   | 0.500000   | 0.000000 | 130985    | 0/0/0       |
| 2026090914 | mid  | central_coordinated        | 0.333333   | 1.000000 | 130985    | 0/0/0       |
| 2026090914 | mid  | shielded_coordinated       | 0.333333   | 1.000000 | 130985    | 0/0/0       |
| 2026090914 | high | desynchronized_independent | 1.000000   | 1.000000 | 126223    | 379/396/678 |
| 2026090914 | high | synchronized_independent   | 1.000000   | 1.000000 | 130941    | 0/0/0       |
| 2026090914 | high | central_coordinated        | 1.000000   | 1.000000 | 130941    | 0/0/0       |
| 2026090914 | high | shielded_coordinated       | 1.000000   | 1.000000 | 130941    | 0/0/0       |
| 2026090915 | low  | desynchronized_independent | 0.333333   | 0.000000 | 131028    | 212/604/213 |
| 2026090915 | low  | synchronized_independent   | 0.244444   | 0.000000 | 131029    | 0/0/0       |
| 2026090915 | low  | central_coordinated        | 0.000000   | –        | 16341     | 0/0/0       |
| 2026090915 | low  | shielded_coordinated       | 0.000000   | –        | 16341     | 0/0/0       |
| 2026090915 | mid  | desynchronized_independent | 0.666667   | 0.505556 | 130985    | 212/604/213 |
| 2026090915 | mid  | synchronized_independent   | 0.500000   | 0.000000 | 130985    | 0/0/0       |
| 2026090915 | mid  | central_coordinated        | 0.333333   | 1.000000 | 130985    | 0/0/0       |
| 2026090915 | mid  | shielded_coordinated       | 0.333333   | 1.000000 | 130985    | 0/0/0       |
| 2026090915 | high | desynchronized_independent | 1.000000   | 1.000000 | 130876    | 212/604/213 |
| 2026090915 | high | synchronized_independent   | 1.000000   | 1.000000 | 130941    | 0/0/0       |
| 2026090915 | high | central_coordinated        | 1.000000   | 1.000000 | 130941    | 0/0/0       |
| 2026090915 | high | shielded_coordinated       | 1.000000   | 1.000000 | 130941    | 0/0/0       |

| Seed       | Cap. | Method                     | Core frac. | On-time  | Max Q (B) | Phases (ms) |
|------------|------|----------------------------|------------|----------|-----------|-------------|
| 2026090916 | low  | desynchronized_independent | 0.333333   | 0.000000 | 131029    | 586/124/90  |
| 2026090916 | low  | synchronized_independent   | 0.244444   | 0.000000 | 131029    | 0/0/0       |
| 2026090916 | low  | central_coordinated        | 0.000000   | –        | 16341     | 0/0/0       |
| 2026090916 | low  | shielded_coordinated       | 0.000000   | –        | 16341     | 0/0/0       |
| 2026090916 | mid  | desynchronized_independent | 0.666667   | 0.505556 | 130985    | 586/124/90  |
| 2026090916 | mid  | synchronized_independent   | 0.500000   | 0.000000 | 130985    | 0/0/0       |
| 2026090916 | mid  | central_coordinated        | 0.333333   | 1.000000 | 130985    | 0/0/0       |
| 2026090916 | mid  | shielded_coordinated       | 0.333333   | 1.000000 | 130985    | 0/0/0       |
| 2026090916 | high | desynchronized_independent | 1.000000   | 1.000000 | 130876    | 586/124/90  |
| 2026090916 | high | synchronized_independent   | 1.000000   | 1.000000 | 130941    | 0/0/0       |
| 2026090916 | high | central_coordinated        | 1.000000   | 1.000000 | 130941    | 0/0/0       |
| 2026090916 | high | shielded_coordinated       | 1.000000   | 1.000000 | 130941    | 0/0/0       |
| 2026090917 | low  | desynchronized_independent | 0.333333   | 0.000000 | 131029    | 13/5/654    |
| 2026090917 | low  | synchronized_independent   | 0.244444   | 0.000000 | 131029    | 0/0/0       |
| 2026090917 | low  | central_coordinated        | 0.000000   | –        | 16341     | 0/0/0       |
| 2026090917 | low  | shielded_coordinated       | 0.000000   | –        | 16341     | 0/0/0       |
| 2026090917 | mid  | desynchronized_independent | 0.666667   | 0.500000 | 130985    | 13/5/654    |
| 2026090917 | mid  | synchronized_independent   | 0.500000   | 0.000000 | 130985    | 0/0/0       |
| 2026090917 | mid  | central_coordinated        | 0.333333   | 1.000000 | 130985    | 0/0/0       |
| 2026090917 | mid  | shielded_coordinated       | 0.333333   | 1.000000 | 130985    | 0/0/0       |
| 2026090917 | high | desynchronized_independent | 1.000000   | 1.000000 | 130876    | 13/5/654    |
| 2026090917 | high | synchronized_independent   | 1.000000   | 1.000000 | 130941    | 0/0/0       |
| 2026090917 | high | central_coordinated        | 1.000000   | 1.000000 | 130941    | 0/0/0       |
| 2026090917 | high | shielded_coordinated       | 1.000000   | 1.000000 | 130941    | 0/0/0       |
| 2026090918 | low  | desynchronized_independent | 0.333333   | 0.000000 | 131029    | 303/966/432 |
| 2026090918 | low  | synchronized_independent   | 0.244444   | 0.000000 | 131029    | 0/0/0       |
| 2026090918 | low  | central_coordinated        | 0.000000   | –        | 16341     | 0/0/0       |
| 2026090918 | low  | shielded_coordinated       | 0.000000   | –        | 16341     | 0/0/0       |
| 2026090918 | mid  | desynchronized_independent | 0.666667   | 0.500000 | 130985    | 303/966/432 |
| 2026090918 | mid  | synchronized_independent   | 0.500000   | 0.000000 | 130985    | 0/0/0       |
| 2026090918 | mid  | central_coordinated        | 0.333333   | 1.000000 | 130985    | 0/0/0       |
| 2026090918 | mid  | shielded_coordinated       | 0.333333   | 1.000000 | 130985    | 0/0/0       |
| 2026090918 | high | desynchronized_independent | 1.000000   | 1.000000 | 91882     | 303/966/432 |
| 2026090918 | high | synchronized_independent   | 1.000000   | 1.000000 | 130941    | 0/0/0       |
| 2026090918 | high | central_coordinated        | 1.000000   | 1.000000 | 130941    | 0/0/0       |
| 2026090918 | high | shielded_coordinated       | 1.000000   | 1.000000 | 130941    | 0/0/0       |
| 2026090919 | low  | desynchronized_independent | 0.333333   | 0.000000 | 131029    | 94/427/937  |
| 2026090919 | low  | synchronized_independent   | 0.244444   | 0.000000 | 131029    | 0/0/0       |
| 2026090919 | low  | central_coordinated        | 0.000000   | –        | 16341     | 0/0/0       |
| 2026090919 | low  | shielded_coordinated       | 0.000000   | –        | 16341     | 0/0/0       |
| 2026090919 | mid  | desynchronized_independent | 0.666667   | 0.500000 | 130985    | 94/427/937  |
| 2026090919 | mid  | synchronized_independent   | 0.500000   | 0.000000 | 130985    | 0/0/0       |
| 2026090919 | mid  | central_coordinated        | 0.333333   | 1.000000 | 130985    | 0/0/0       |
| 2026090919 | mid  | shielded_coordinated       | 0.333333   | 1.000000 | 130985    | 0/0/0       |
| 2026090919 | high | desynchronized_independent | 1.000000   | 1.000000 | 93848     | 94/427/937  |
| 2026090919 | high | synchronized_independent   | 1.000000   | 1.000000 | 130941    | 0/0/0       |
| 2026090919 | high | central_coordinated        | 1.000000   | 1.000000 | 130941    | 0/0/0       |
| 2026090919 | high | shielded_coordinated       | 1.000000   | 1.000000 | 130941    | 0/0/0       |
| 2026090920 | low  | desynchronized_independent | 0.333333   | 0.000000 | 131029    | 699/148/926 |
| 2026090920 | low  | synchronized_independent   | 0.244444   | 0.000000 | 131029    | 0/0/0       |
| 2026090920 | low  | central_coordinated        | 0.000000   | –        | 16341     | 0/0/0       |
| 2026090920 | low  | shielded_coordinated       | 0.000000   | –        | 16341     | 0/0/0       |
| 2026090920 | mid  | desynchronized_independent | 0.666667   | 0.505556 | 130985    | 699/148/926 |
| 2026090920 | mid  | synchronized_independent   | 0.500000   | 0.000000 | 130985    | 0/0/0       |
| 2026090920 | mid  | central_coordinated        | 0.333333   | 1.000000 | 130985    | 0/0/0       |
| 2026090920 | mid  | shielded_coordinated       | 0.333333   | 1.000000 | 130985    | 0/0/0       |
| 2026090920 | high | desynchronized_independent | 1.000000   | 1.000000 | 72483     | 699/148/926 |
| 2026090920 | high | synchronized_independent   | 1.000000   | 1.000000 | 130941    | 0/0/0       |
| 2026090920 | high | central_coordinated        | 1.000000   | 1.000000 | 130941    | 0/0/0       |
| 2026090920 | high | shielded_coordinated       | 1.000000   | 1.000000 | 130941    | 0/0/0       |

## VII. COMPLETE SCHEDULE-ANCHOR RECONSTRUCTION

Post-release latency is result return minus actual event creation. Nominal latency is result return minus scheduled release. The 1-s quantity in the main paper is a post-release service SLO, not a complete schedule-to-result deadline.

TABLE V: Complete accepted-hardware timing reconstruction (648 retained cells; latency in ms).

| Session | Method          | Cap. | Dev. | $n_c$ | Post   | Nom.   | Late p50 | p95    | p99    |
|---------|-----------------|------|------|-------|--------|--------|----------|--------|--------|
| F12_S01 | fixed_local     | low  | P1   | 0     | –      | –      | 1862.2   | 1920.9 | 1924.1 |
| F12_S01 | fixed_local     | low  | P2   | 0     | –      | –      | 760.3    | 831.8  | 834.1  |
| F12_S01 | fixed_local     | low  | P3   | 0     | –      | –      | 86.3     | 155.7  | 162.0  |
| F12_S01 | fixed_local     | mid  | P1   | 0     | –      | –      | 1828.4   | 1979.1 | 1980.6 |
| F12_S01 | fixed_local     | mid  | P2   | 0     | –      | –      | 636.8    | 775.7  | 780.4  |
| F12_S01 | fixed_local     | mid  | P3   | 0     | –      | –      | 87.5     | 165.1  | 172.5  |
| F12_S01 | fixed_local     | high | P1   | 0     | –      | –      | 1827.5   | 1863.0 | 1913.4 |
| F12_S01 | fixed_local     | high | P2   | 0     | –      | –      | 637.0    | 671.0  | 673.3  |
| F12_S01 | fixed_local     | high | P3   | 0     | –      | –      | 84.4     | 156.5  | 163.5  |
| F12_S01 | fixed_summary   | low  | P1   | 0     | –      | –      | 1952.6   | 2002.0 | 2006.5 |
| F12_S01 | fixed_summary   | low  | P2   | 0     | –      | –      | 712.1    | 750.1  | 753.4  |
| F12_S01 | fixed_summary   | low  | P3   | 0     | –      | –      | 84.0     | 156.4  | 163.8  |
| F12_S01 | fixed_summary   | mid  | P1   | 0     | –      | –      | 1922.6   | 1997.1 | 2001.5 |
| F12_S01 | fixed_summary   | mid  | P2   | 0     | –      | –      | 716.0    | 804.2  | 806.5  |
| F12_S01 | fixed_summary   | mid  | P3   | 0     | –      | –      | 85.2     | 153.6  | 160.0  |
| F12_S01 | fixed_summary   | high | P1   | 0     | –      | –      | 1925.6   | 1978.0 | 1981.4 |
| F12_S01 | fixed_summary   | high | P2   | 0     | –      | –      | 745.5    | 802.9  | 806.8  |
| F12_S01 | fixed_summary   | high | P3   | 0     | –      | –      | 88.3     | 159.6  | 166.3  |
| F12_S01 | fixed_full      | low  | P1   | 90    | 0.0000 | 0.0000 | 1887.7   | 1981.0 | 1983.3 |
| F12_S01 | fixed_full      | low  | P2   | 90    | 0.0000 | 0.0000 | 736.5    | 827.9  | 830.4  |
| F12_S01 | fixed_full      | low  | P3   | 90    | 0.0000 | 0.0000 | 79.3     | 139.5  | 145.3  |
| F12_S01 | fixed_full      | mid  | P1   | 90    | 0.0000 | 0.0000 | 1859.1   | 2483.1 | 2486.7 |
| F12_S01 | fixed_full      | mid  | P2   | 90    | 0.0000 | 0.0000 | 686.7    | 1341.8 | 1346.6 |
| F12_S01 | fixed_full      | mid  | P3   | 90    | 0.0000 | 0.0000 | 79.2     | 141.9  | 148.5  |
| F12_S01 | fixed_full      | high | P1   | 90    | 0.9667 | 0.0000 | 2024.3   | 2101.8 | 2105.2 |
| F12_S01 | fixed_full      | high | P2   | 90    | 0.9222 | 0.0000 | 830.5    | 948.8  | 952.3  |
| F12_S01 | fixed_full      | high | P3   | 90    | 1.0000 | 1.0000 | 78.6     | 141.2  | 145.6  |
| F12_S01 | independent_dpp | low  | P1   | 22    | 0.0000 | 0.0000 | 1851.5   | 2183.8 | 2301.2 |
| F12_S01 | independent_dpp | low  | P2   | 22    | 0.0000 | 0.0000 | 661.8    | 792.4  | 1053.9 |
| F12_S01 | independent_dpp | low  | P3   | 22    | 0.0000 | 0.0000 | 82.3     | 152.1  | 156.3  |
| F12_S01 | independent_dpp | mid  | P1   | 45    | 0.0000 | 0.0000 | 1998.0   | 2108.8 | 2111.0 |
| F12_S01 | independent_dpp | mid  | P2   | 45    | 0.0000 | 0.0000 | 754.9    | 822.0  | 824.8  |
| F12_S01 | independent_dpp | mid  | P3   | 45    | 0.0000 | 0.0000 | 81.4     | 146.9  | 152.4  |
| F12_S01 | independent_dpp | high | P1   | 90    | 0.9444 | 0.0000 | 1967.9   | 2006.6 | 2009.6 |
| F12_S01 | independent_dpp | high | P2   | 90    | 0.9556 | 0.0000 | 797.6    | 836.1  | 841.8  |
| F12_S01 | independent_dpp | high | P3   | 90    | 1.0000 | 1.0000 | 78.4     | 136.1  | 140.8  |
| F12_S01 | central_dpp     | low  | P1   | 0     | –      | –      | 2049.7   | 2151.4 | 2153.5 |
| F12_S01 | central_dpp     | low  | P2   | 0     | –      | –      | 858.4    | 942.6  | 947.1  |
| F12_S01 | central_dpp     | low  | P3   | 0     | –      | –      | 83.2     | 148.0  | 154.0  |
| F12_S01 | central_dpp     | mid  | P1   | 30    | 0.8000 | 0.0000 | 1910.4   | 1992.9 | 1995.4 |
| F12_S01 | central_dpp     | mid  | P2   | 30    | 0.8000 | 0.0000 | 732.4    | 809.6  | 813.0  |
| F12_S01 | central_dpp     | mid  | P3   | 30    | 1.0000 | 1.0000 | 83.1     | 152.5  | 159.3  |
| F12_S01 | central_dpp     | high | P1   | 90    | 1.0000 | 0.0000 | 2070.5   | 2105.4 | 2109.1 |
| F12_S01 | central_dpp     | high | P2   | 90    | 0.9889 | 0.0000 | 907.5    | 949.5  | 953.2  |
| F12_S01 | central_dpp     | high | P3   | 90    | 1.0000 | 1.0000 | 78.5     | 144.9  | 150.8  |
| F12_S01 | shielded_dpp    | low  | P1   | 0     | –      | –      | 1771.7   | 1832.4 | 1835.5 |
| F12_S01 | shielded_dpp    | low  | P2   | 0     | –      | –      | 678.1    | 749.6  | 786.5  |
| F12_S01 | shielded_dpp    | low  | P3   | 0     | –      | –      | 87.8     | 156.7  | 162.3  |
| F12_S01 | shielded_dpp    | mid  | P1   | 30    | 0.9000 | 0.0000 | 1934.5   | 2160.6 | 2163.7 |
| F12_S01 | shielded_dpp    | mid  | P2   | 30    | 1.0000 | 0.0000 | 856.4    | 964.9  | 967.9  |
| F12_S01 | shielded_dpp    | mid  | P3   | 30    | 1.0000 | 1.0000 | 86.3     | 149.8  | 155.9  |
| F12_S01 | shielded_dpp    | high | P1   | 90    | 0.9333 | 0.0000 | 1979.2   | 2065.8 | 2070.3 |
| F12_S01 | shielded_dpp    | high | P2   | 90    | 0.9444 | 0.0000 | 876.4    | 913.1  | 974.2  |
| F12_S01 | shielded_dpp    | high | P3   | 90    | 1.0000 | 1.0000 | 79.7     | 136.0  | 140.8  |
| F12_S02 | fixed_local     | low  | P1   | 0     | –      | –      | 1886.4   | 2021.6 | 2024.1 |
| F12_S02 | fixed_local     | low  | P2   | 0     | –      | –      | 594.0    | 740.2  | 743.2  |
| F12_S02 | fixed_local     | low  | P3   | 0     | –      | –      | 88.2     | 164.6  | 171.5  |
| F12_S02 | fixed_local     | mid  | P1   | 0     | –      | –      | 2003.6   | 2046.7 | 2051.4 |
| F12_S02 | fixed_local     | mid  | P2   | 0     | –      | –      | 717.1    | 761.0  | 764.1  |
| F12_S02 | fixed_local     | mid  | P3   | 0     | –      | –      | 92.2     | 161.7  | 168.4  |
| F12_S02 | fixed_local     | high | P1   | 0     | –      | –      | 1896.5   | 1938.2 | 1941.3 |
| F12_S02 | fixed_local     | high | P2   | 0     | –      | –      | 737.1    | 775.7  | 778.9  |
| F12_S02 | fixed_local     | high | P3   | 0     | –      | –      | 91.3     | 160.5  | 167.6  |
| F12_S02 | fixed_summary   | low  | P1   | 0     | –      | –      | 1725.1   | 1764.9 | 1767.8 |
| F12_S02 | fixed_summary   | low  | P2   | 0     | –      | –      | 550.6    | 585.7  | 589.6  |
| F12_S02 | fixed_summary   | low  | P3   | 0     | –      | –      | 90.6     | 160.8  | 167.3  |
| F12_S02 | fixed_summary   | mid  | P1   | 0     | –      | –      | 1915.3   | 2065.5 | 2072.1 |
| F12_S02 | fixed_summary   | mid  | P2   | 0     | –      | –      | 677.3    | 1103.5 | 1106.6 |
| F12_S02 | fixed_summary   | mid  | P3   | 0     | –      | –      | 85.8     | 156.6  | 162.7  |
| F12_S02 | fixed_summary   | high | P1   | 0     | –      | –      | 1878.3   | 1949.3 | 1959.8 |
| F12_S02 | fixed_summary   | high | P2   | 0     | –      | –      | 713.1    | 787.0  | 798.8  |
| F12_S02 | fixed_summary   | high | P3   | 0     | –      | –      | 89.0     | 158.6  | 165.8  |
| F12_S02 | fixed_full      | low  | P1   | 90    | 0.0000 | 0.0000 | 1929.0   | 2110.4 | 2114.6 |
| F12_S02 | fixed_full      | low  | P2   | 90    | 0.0000 | 0.0000 | 763.0    | 876.5  | 879.5  |

| Session | Method          | Cap. | Dev. | $n_c$ | Post   | Nom.   | Late p50 | p95    | p99    |
|---------|-----------------|------|------|-------|--------|--------|----------|--------|--------|
| F12_S02 | fixed_full      | low  | P3   | 90    | 0.0000 | 0.0000 | 78.2     | 135.9  | 142.5  |
| F12_S02 | fixed_full      | mid  | P1   | 90    | 0.0000 | 0.0000 | 2251.6   | 2305.8 | 2309.9 |
| F12_S02 | fixed_full      | mid  | P2   | 90    | 0.0000 | 0.0000 | 616.5    | 733.8  | 750.5  |
| F12_S02 | fixed_full      | mid  | P3   | 90    | 0.0000 | 0.0000 | 76.4     | 136.6  | 142.5  |
| F12_S02 | fixed_full      | high | P1   | 90    | 1.0000 | 0.0000 | 1969.5   | 1996.6 | 2001.5 |
| F12_S02 | fixed_full      | high | P2   | 90    | 0.9889 | 0.0000 | 827.2    | 865.1  | 867.9  |
| F12_S02 | fixed_full      | high | P3   | 90    | 1.0000 | 1.0000 | 80.0     | 145.5  | 151.7  |
| F12_S02 | independent_dpp | low  | P1   | 22    | 0.0000 | 0.0000 | 1906.4   | 2060.6 | 2064.0 |
| F12_S02 | independent_dpp | low  | P2   | 22    | 0.0000 | 0.0000 | 748.0    | 899.5  | 904.3  |
| F12_S02 | independent_dpp | low  | P3   | 22    | 0.0000 | 0.0000 | 78.9     | 142.0  | 147.5  |
| F12_S02 | independent_dpp | mid  | P1   | 45    | 0.0000 | 0.0000 | 1980.3   | 2013.9 | 2015.6 |
| F12_S02 | independent_dpp | mid  | P2   | 45    | 0.0000 | 0.0000 | 703.8    | 732.1  | 734.0  |
| F12_S02 | independent_dpp | mid  | P3   | 45    | 0.0000 | 0.0000 | 79.5     | 147.8  | 152.5  |
| F12_S02 | independent_dpp | high | P1   | 90    | 0.9111 | 0.0000 | 1982.0   | 2052.8 | 2056.0 |
| F12_S02 | independent_dpp | high | P2   | 90    | 0.9333 | 0.0000 | 787.2    | 831.0  | 832.9  |
| F12_S02 | independent_dpp | high | P3   | 90    | 1.0000 | 1.0000 | 78.0     | 137.5  | 141.7  |
| F12_S02 | central_dpp     | low  | P1   | 0     | –      | –      | 2017.6   | 2154.2 | 2157.5 |
| F12_S02 | central_dpp     | low  | P2   | 0     | –      | –      | 738.5    | 861.7  | 865.2  |
| F12_S02 | central_dpp     | low  | P3   | 0     | –      | –      | 87.9     | 158.3  | 164.1  |
| F12_S02 | central_dpp     | mid  | P1   | 30    | 0.7333 | 0.0000 | 1936.7   | 2071.6 | 2074.8 |
| F12_S02 | central_dpp     | mid  | P2   | 30    | 0.9333 | 0.0000 | 881.1    | 913.6  | 916.2  |
| F12_S02 | central_dpp     | mid  | P3   | 30    | 1.0000 | 1.0000 | 86.2     | 150.4  | 156.4  |
| F12_S02 | central_dpp     | high | P1   | 90    | 0.9222 | 0.0000 | 1942.0   | 1995.4 | 2000.0 |
| F12_S02 | central_dpp     | high | P2   | 90    | 0.9444 | 0.0000 | 816.7    | 862.3  | 865.6  |
| F12_S02 | central_dpp     | high | P3   | 90    | 1.0000 | 1.0000 | 81.3     | 145.8  | 151.8  |
| F12_S02 | shielded_dpp    | low  | P1   | 0     | –      | –      | 1931.4   | 1995.7 | 1999.1 |
| F12_S02 | shielded_dpp    | low  | P2   | 0     | –      | –      | 842.4    | 903.9  | 908.0  |
| F12_S02 | shielded_dpp    | low  | P3   | 0     | –      | –      | 86.6     | 160.6  | 168.3  |
| F12_S02 | shielded_dpp    | mid  | P1   | 30    | 0.9000 | 0.0000 | 2120.6   | 2150.6 | 2151.7 |
| F12_S02 | shielded_dpp    | mid  | P2   | 30    | 0.9000 | 0.0000 | 793.9    | 826.4  | 828.9  |
| F12_S02 | shielded_dpp    | mid  | P3   | 30    | 1.0000 | 1.0000 | 80.3     | 148.0  | 155.9  |
| F12_S02 | shielded_dpp    | high | P1   | 90    | 0.9444 | 0.0000 | 1954.9   | 2007.4 | 2039.4 |
| F12_S02 | shielded_dpp    | high | P2   | 90    | 0.9333 | 0.0000 | 781.0    | 858.6  | 873.6  |
| F12_S02 | shielded_dpp    | high | P3   | 90    | 1.0000 | 1.0000 | 81.5     | 140.4  | 145.8  |
| F12_S03 | fixed_local     | low  | P1   | 0     | –      | –      | 1892.3   | 1942.8 | 1950.8 |
| F12_S03 | fixed_local     | low  | P2   | 0     | –      | –      | 676.8    | 732.8  | 744.0  |
| F12_S03 | fixed_local     | low  | P3   | 0     | –      | –      | 90.0     | 162.6  | 169.3  |
| F12_S03 | fixed_local     | mid  | P1   | 0     | –      | –      | 1823.3   | 1868.5 | 1871.1 |
| F12_S03 | fixed_local     | mid  | P2   | 0     | –      | –      | 736.6    | 837.7  | 840.2  |
| F12_S03 | fixed_local     | mid  | P3   | 0     | –      | –      | 89.2     | 160.3  | 166.4  |
| F12_S03 | fixed_local     | high | P1   | 0     | –      | –      | 1939.5   | 2052.7 | 2061.3 |
| F12_S03 | fixed_local     | high | P2   | 0     | –      | –      | 724.4    | 838.4  | 852.7  |
| F12_S03 | fixed_local     | high | P3   | 0     | –      | –      | 90.8     | 167.3  | 174.3  |
| F12_S03 | fixed_summary   | low  | P1   | 0     | –      | –      | 1926.4   | 1967.6 | 1971.1 |
| F12_S03 | fixed_summary   | low  | P2   | 0     | –      | –      | 710.5    | 757.2  | 762.6  |
| F12_S03 | fixed_summary   | low  | P3   | 0     | –      | –      | 90.1     | 166.0  | 172.4  |
| F12_S03 | fixed_summary   | mid  | P1   | 0     | –      | –      | 2190.5   | 2234.0 | 2237.6 |
| F12_S03 | fixed_summary   | mid  | P2   | 0     | –      | –      | 877.1    | 918.5  | 922.6  |
| F12_S03 | fixed_summary   | mid  | P3   | 0     | –      | –      | 90.8     | 162.8  | 169.3  |
| F12_S03 | fixed_summary   | high | P1   | 0     | –      | –      | 2024.1   | 2062.5 | 2066.0 |
| F12_S03 | fixed_summary   | high | P2   | 0     | –      | –      | 748.8    | 787.7  | 790.7  |
| F12_S03 | fixed_summary   | high | P3   | 0     | –      | –      | 89.3     | 160.2  | 166.3  |
| F12_S03 | fixed_full      | low  | P1   | 90    | 0.0000 | 0.0000 | 1963.8   | 2094.5 | 2097.1 |
| F12_S03 | fixed_full      | low  | P2   | 90    | 0.0000 | 0.0000 | 864.3    | 926.3  | 947.9  |
| F12_S03 | fixed_full      | low  | P3   | 90    | 0.0000 | 0.0000 | 78.2     | 147.4  | 153.9  |
| F12_S03 | fixed_full      | mid  | P1   | 90    | 0.0000 | 0.0000 | 1903.6   | 1975.4 | 1976.5 |
| F12_S03 | fixed_full      | mid  | P2   | 90    | 0.0000 | 0.0000 | 752.0    | 824.8  | 829.8  |
| F12_S03 | fixed_full      | mid  | P3   | 90    | 0.0000 | 0.0000 | 80.7     | 141.3  | 147.6  |
| F12_S03 | fixed_full      | high | P1   | 90    | 0.9667 | 0.0000 | 2006.1   | 2050.0 | 2098.4 |
| F12_S03 | fixed_full      | high | P2   | 90    | 0.9333 | 0.0000 | 847.8    | 892.2  | 1004.3 |
| F12_S03 | fixed_full      | high | P3   | 90    | 1.0000 | 1.0000 | 86.9     | 150.9  | 156.0  |
| F12_S03 | independent_dpp | low  | P1   | 22    | 0.0000 | 0.0000 | 1937.5   | 2023.7 | 2025.5 |
| F12_S03 | independent_dpp | low  | P2   | 22    | 0.0000 | 0.0000 | 755.3    | 832.0  | 841.6  |
| F12_S03 | independent_dpp | low  | P3   | 22    | 0.0000 | 0.0000 | 82.0     | 151.6  | 158.6  |
| F12_S03 | independent_dpp | mid  | P1   | 45    | 0.0000 | 0.0000 | 7335.7   | 7390.7 | 7395.4 |
| F12_S03 | independent_dpp | mid  | P2   | 45    | 0.0000 | 0.0000 | 6236.0   | 6290.3 | 6295.7 |
| F12_S03 | independent_dpp | mid  | P3   | 45    | 0.0000 | 0.0000 | 4851.8   | 4918.9 | 4924.2 |
| F12_S03 | independent_dpp | high | P1   | 90    | 0.9889 | 0.0000 | 1867.0   | 2014.7 | 2018.8 |
| F12_S03 | independent_dpp | high | P2   | 90    | 0.9556 | 0.0000 | 699.5    | 814.5  | 817.0  |
| F12_S03 | independent_dpp | high | P3   | 90    | 1.0000 | 1.0000 | 74.3     | 134.1  | 139.7  |
| F12_S03 | central_dpp     | low  | P1   | 0     | –      | –      | 1984.5   | 2024.7 | 2030.2 |
| F12_S03 | central_dpp     | low  | P2   | 0     | –      | –      | 838.4    | 875.1  | 878.3  |
| F12_S03 | central_dpp     | low  | P3   | 0     | –      | –      | 90.1     | 161.4  | 168.3  |
| F12_S03 | central_dpp     | mid  | P1   | 30    | 0.8667 | 0.0000 | 1949.1   | 2028.9 | 2031.1 |
| F12_S03 | central_dpp     | mid  | P2   | 30    | 0.7667 | 0.0000 | 915.3    | 954.2  | 957.1  |
| F12_S03 | central_dpp     | mid  | P3   | 30    | 1.0000 | 1.0000 | 81.0     | 147.8  | 154.9  |
| F12_S03 | central_dpp     | high | P1   | 90    | 0.9667 | 0.0000 | 1963.1   | 2023.1 | 2025.9 |
| F12_S03 | central_dpp     | high | P2   | 90    | 1.0000 | 0.0000 | 856.1    | 892.5  | 896.6  |
| F12_S03 | central_dpp     | high | P3   | 90    | 1.0000 | 1.0000 | 73.2     | 133.8  | 140.1  |
| F12_S03 | shielded_dpp    | low  | P1   | 0     | –      | –      | 2074.6   | 2116.7 | 2120.3 |
| F12_S03 | shielded_dpp    | low  | P2   | 0     | –      | –      | 627.9    | 665.5  | 668.1  |
| F12_S03 | shielded_dpp    | low  | P3   | 0     | –      | –      | 84.1     | 154.8  | 160.5  |
| F12_S03 | shielded_dpp    | mid  | P1   | 30    | 0.8000 | 0.0000 | 2003.3   | 2147.9 | 2150.1 |

| Session | Method          | Cap. | Dev. | $n_c$ | Post   | Nom.   | Late p50 | p95    | p99    |
|---------|-----------------|------|------|-------|--------|--------|----------|--------|--------|
| F12_S03 | shielded_dpp    | mid  | P2   | 30    | 0.8333 | 0.0000 | 758.1    | 862.1  | 892.8  |
| F12_S03 | shielded_dpp    | mid  | P3   | 30    | 1.0000 | 1.0000 | 86.6     | 151.2  | 156.3  |
| F12_S03 | shielded_dpp    | high | P1   | 90    | 0.9333 | 0.0000 | 2057.5   | 2130.4 | 2139.2 |
| F12_S03 | shielded_dpp    | high | P2   | 90    | 0.9778 | 0.0000 | 939.6    | 989.1  | 992.6  |
| F12_S03 | shielded_dpp    | high | P3   | 90    | 1.0000 | 1.0000 | 77.9     | 142.0  | 146.9  |
| F12_S04 | fixed_local     | low  | P1   | 0     | –      | –      | 1879.6   | 1975.4 | 1977.9 |
| F12_S04 | fixed_local     | low  | P2   | 0     | –      | –      | 657.9    | 744.4  | 746.8  |
| F12_S04 | fixed_local     | low  | P3   | 0     | –      | –      | 91.4     | 163.7  | 170.4  |
| F12_S04 | fixed_local     | mid  | P1   | 0     | –      | –      | 1944.5   | 2017.8 | 2022.3 |
| F12_S04 | fixed_local     | mid  | P2   | 0     | –      | –      | 724.9    | 795.4  | 798.7  |
| F12_S04 | fixed_local     | mid  | P3   | 0     | –      | –      | 95.3     | 169.4  | 176.3  |
| F12_S04 | fixed_local     | high | P1   | 0     | –      | –      | 1896.6   | 1934.6 | 1938.1 |
| F12_S04 | fixed_local     | high | P2   | 0     | –      | –      | 743.8    | 780.2  | 783.8  |
| F12_S04 | fixed_local     | high | P3   | 0     | –      | –      | 87.9     | 160.4  | 166.0  |
| F12_S04 | fixed_summary   | low  | P1   | 0     | –      | –      | 1958.8   | 2049.4 | 2059.2 |
| F12_S04 | fixed_summary   | low  | P2   | 0     | –      | –      | 719.0    | 800.6  | 806.0  |
| F12_S04 | fixed_summary   | low  | P3   | 0     | –      | –      | 87.6     | 156.2  | 162.2  |
| F12_S04 | fixed_summary   | mid  | P1   | 0     | –      | –      | 1929.8   | 2022.4 | 2027.0 |
| F12_S04 | fixed_summary   | mid  | P2   | 0     | –      | –      | 706.0    | 830.5  | 834.7  |
| F12_S04 | fixed_summary   | mid  | P3   | 0     | –      | –      | 89.1     | 158.3  | 164.2  |
| F12_S04 | fixed_summary   | high | P1   | 0     | –      | –      | 1973.7   | 2015.9 | 2019.2 |
| F12_S04 | fixed_summary   | high | P2   | 0     | –      | –      | 595.9    | 629.4  | 630.5  |
| F12_S04 | fixed_summary   | high | P3   | 0     | –      | –      | 88.2     | 155.6  | 161.3  |
| F12_S04 | fixed_full      | low  | P1   | 90    | 0.0000 | 0.0000 | 2135.3   | 2299.8 | 2301.8 |
| F12_S04 | fixed_full      | low  | P2   | 90    | 0.0000 | 0.0000 | 843.4    | 1033.6 | 1036.2 |
| F12_S04 | fixed_full      | low  | P3   | 90    | 0.0000 | 0.0000 | 78.4     | 140.7  | 146.6  |
| F12_S04 | fixed_full      | mid  | P1   | 90    | 0.0000 | 0.0000 | 1990.7   | 2057.3 | 2060.3 |
| F12_S04 | fixed_full      | mid  | P2   | 90    | 0.0000 | 0.0000 | 864.7    | 897.3  | 900.8  |
| F12_S04 | fixed_full      | mid  | P3   | 90    | 0.0000 | 0.0000 | 78.5     | 141.6  | 146.8  |
| F12_S04 | fixed_full      | high | P1   | 90    | 0.9556 | 0.0000 | 1968.2   | 2168.1 | 2172.8 |
| F12_S04 | fixed_full      | high | P2   | 90    | 0.9222 | 0.0000 | 746.4    | 946.3  | 950.5  |
| F12_S04 | fixed_full      | high | P3   | 90    | 1.0000 | 1.0000 | 78.6     | 142.2  | 146.8  |
| F12_S04 | independent_dpp | low  | P1   | 22    | 0.0000 | 0.0000 | 2113.4   | 2196.6 | 2199.2 |
| F12_S04 | independent_dpp | low  | P2   | 22    | 0.0000 | 0.0000 | 1004.2   | 1036.6 | 1038.8 |
| F12_S04 | independent_dpp | low  | P3   | 22    | 0.0000 | 0.0000 | 85.1     | 154.4  | 159.8  |
| F12_S04 | independent_dpp | mid  | P1   | 45    | 0.0000 | 0.0000 | 1990.7   | 2228.1 | 2232.6 |
| F12_S04 | independent_dpp | mid  | P2   | 45    | 0.0000 | 0.0000 | 837.2    | 975.7  | 977.9  |
| F12_S04 | independent_dpp | mid  | P3   | 45    | 0.0000 | 0.0000 | 77.9     | 143.7  | 149.7  |
| F12_S04 | independent_dpp | high | P1   | 90    | 0.9444 | 0.0000 | 2007.0   | 2091.5 | 2094.1 |
| F12_S04 | independent_dpp | high | P2   | 90    | 0.9556 | 0.0000 | 806.0    | 927.9  | 929.9  |
| F12_S04 | independent_dpp | high | P3   | 90    | 1.0000 | 1.0000 | 80.4     | 145.8  | 151.8  |
| F12_S04 | central_dpp     | low  | P1   | 0     | –      | –      | 1985.7   | 2026.6 | 2029.4 |
| F12_S04 | central_dpp     | low  | P2   | 0     | –      | –      | 669.0    | 706.4  | 709.7  |
| F12_S04 | central_dpp     | low  | P3   | 0     | –      | –      | 85.8     | 158.2  | 163.9  |
| F12_S04 | central_dpp     | mid  | P1   | 30    | 0.6667 | 0.0000 | 1908.9   | 1971.8 | 1982.7 |
| F12_S04 | central_dpp     | mid  | P2   | 30    | 0.7667 | 0.0000 | 747.3    | 842.3  | 846.1  |
| F12_S04 | central_dpp     | mid  | P3   | 30    | 1.0000 | 1.0000 | 86.9     | 156.1  | 162.2  |
| F12_S04 | central_dpp     | high | P1   | 90    | 0.9556 | 0.0000 | 2089.1   | 2217.9 | 2220.9 |
| F12_S04 | central_dpp     | high | P2   | 90    | 0.9889 | 0.0000 | 855.7    | 922.1  | 923.9  |
| F12_S04 | central_dpp     | high | P3   | 90    | 1.0000 | 1.0000 | 74.0     | 133.8  | 139.0  |
| F12_S04 | shielded_dpp    | low  | P1   | 0     | –      | –      | 1873.0   | 1929.6 | 1934.2 |
| F12_S04 | shielded_dpp    | low  | P2   | 0     | –      | –      | 648.1    | 713.7  | 716.7  |
| F12_S04 | shielded_dpp    | low  | P3   | 0     | –      | –      | 85.0     | 156.3  | 162.7  |
| F12_S04 | shielded_dpp    | mid  | P1   | 30    | 0.7667 | 0.0000 | 1978.4   | 2044.0 | 2047.9 |
| F12_S04 | shielded_dpp    | mid  | P2   | 30    | 0.7667 | 0.0000 | 795.4    | 884.6  | 887.1  |
| F12_S04 | shielded_dpp    | mid  | P3   | 30    | 1.0000 | 1.0000 | 81.0     | 144.6  | 150.5  |
| F12_S04 | shielded_dpp    | high | P1   | 90    | 0.9778 | 0.0000 | 1987.3   | 2025.7 | 2029.0 |
| F12_S04 | shielded_dpp    | high | P2   | 90    | 0.9111 | 0.0000 | 724.4    | 854.0  | 858.5  |
| F12_S04 | shielded_dpp    | high | P3   | 90    | 1.0000 | 1.0000 | 78.4     | 144.4  | 150.5  |
| F12_S05 | fixed_local     | low  | P1   | 0     | –      | –      | 2011.8   | 2048.3 | 2050.5 |
| F12_S05 | fixed_local     | low  | P2   | 0     | –      | –      | 735.0    | 769.7  | 771.7  |
| F12_S05 | fixed_local     | low  | P3   | 0     | –      | –      | 88.0     | 158.3  | 164.5  |
| F12_S05 | fixed_local     | mid  | P1   | 0     | –      | –      | 1826.1   | 1896.2 | 1901.4 |
| F12_S05 | fixed_local     | mid  | P2   | 0     | –      | –      | 665.2    | 704.7  | 706.8  |
| F12_S05 | fixed_local     | mid  | P3   | 0     | –      | –      | 86.6     | 158.2  | 164.3  |
| F12_S05 | fixed_local     | high | P1   | 0     | –      | –      | 1917.4   | 1963.7 | 1966.4 |
| F12_S05 | fixed_local     | high | P2   | 0     | –      | –      | 763.8    | 800.2  | 802.8  |
| F12_S05 | fixed_local     | high | P3   | 0     | –      | –      | 90.4     | 169.1  | 175.5  |
| F12_S05 | fixed_summary   | low  | P1   | 0     | –      | –      | 1836.5   | 1895.6 | 1899.3 |
| F12_S05 | fixed_summary   | low  | P2   | 0     | –      | –      | 624.0    | 695.4  | 698.3  |
| F12_S05 | fixed_summary   | low  | P3   | 0     | –      | –      | 88.9     | 160.6  | 166.5  |
| F12_S05 | fixed_summary   | mid  | P1   | 0     | –      | –      | 1787.3   | 1904.6 | 1908.2 |
| F12_S05 | fixed_summary   | mid  | P2   | 0     | –      | –      | 749.4    | 864.3  | 868.8  |
| F12_S05 | fixed_summary   | mid  | P3   | 0     | –      | –      | 91.0     | 163.6  | 170.1  |
| F12_S05 | fixed_summary   | high | P1   | 0     | –      | –      | 1869.3   | 1931.6 | 1935.0 |
| F12_S05 | fixed_summary   | high | P2   | 0     | –      | –      | 712.6    | 774.4  | 779.4  |
| F12_S05 | fixed_summary   | high | P3   | 0     | –      | –      | 86.0     | 152.7  | 159.9  |
| F12_S05 | fixed_full      | low  | P1   | 90    | 0.0000 | 0.0000 | 2017.7   | 2049.2 | 2051.7 |
| F12_S05 | fixed_full      | low  | P2   | 90    | 0.0000 | 0.0000 | 779.2    | 840.4  | 844.0  |
| F12_S05 | fixed_full      | low  | P3   | 90    | 0.0000 | 0.0000 | 76.6     | 140.7  | 146.2  |
| F12_S05 | fixed_full      | mid  | P1   | 90    | 0.0000 | 0.0000 | 1920.6   | 2001.0 | 2049.4 |
| F12_S05 | fixed_full      | mid  | P2   | 90    | 0.0000 | 0.0000 | 746.0    | 797.5  | 843.3  |
| F12_S05 | fixed_full      | mid  | P3   | 90    | 0.0000 | 0.0000 | 78.4     | 144.6  | 151.0  |

| Session | Method          | Cap. | Dev. | $n_c$ | Post   | Nom.   | Late p50 | p95    | p99    |
|---------|-----------------|------|------|-------|--------|--------|----------|--------|--------|
| F12_S05 | fixed_full      | high | P1   | 90    | 0.9111 | 0.0000 | 1861.7   | 1922.9 | 1926.9 |
| F12_S05 | fixed_full      | high | P2   | 90    | 0.9222 | 0.0000 | 777.4    | 829.5  | 831.5  |
| F12_S05 | fixed_full      | high | P3   | 90    | 1.0000 | 1.0000 | 81.1     | 140.9  | 146.8  |
| F12_S05 | independent_dpp | low  | P1   | 22    | 0.0000 | 0.0000 | 1901.3   | 1976.4 | 1981.7 |
| F12_S05 | independent_dpp | low  | P2   | 22    | 0.0000 | 0.0000 | 654.1    | 712.2  | 715.8  |
| F12_S05 | independent_dpp | low  | P3   | 22    | 0.0000 | 0.0000 | 81.8     | 145.1  | 151.4  |
| F12_S05 | independent_dpp | mid  | P1   | 45    | 0.0000 | 0.0000 | 1841.5   | 1921.5 | 1925.4 |
| F12_S05 | independent_dpp | mid  | P2   | 45    | 0.0000 | 0.0000 | 675.7    | 737.0  | 747.7  |
| F12_S05 | independent_dpp | mid  | P3   | 45    | 0.0000 | 0.0000 | 74.9     | 137.5  | 142.5  |
| F12_S05 | independent_dpp | high | P1   | 90    | 0.9778 | 0.0000 | 1962.0   | 2083.5 | 2087.9 |
| F12_S05 | independent_dpp | high | P2   | 90    | 0.9778 | 0.0000 | 886.6    | 920.4  | 923.2  |
| F12_S05 | independent_dpp | high | P3   | 90    | 1.0000 | 1.0000 | 76.8     | 139.6  | 145.5  |
| F12_S05 | central_dpp     | low  | P1   | 0     | –      | –      | 1957.7   | 1999.1 | 2002.0 |
| F12_S05 | central_dpp     | low  | P2   | 0     | –      | –      | 769.8    | 811.2  | 814.7  |
| F12_S05 | central_dpp     | low  | P3   | 0     | –      | –      | 85.3     | 153.6  | 160.1  |
| F12_S05 | central_dpp     | mid  | P1   | 30    | 1.0000 | 0.0000 | 1912.7   | 1940.1 | 1942.0 |
| F12_S05 | central_dpp     | mid  | P2   | 30    | 0.8000 | 0.0000 | 793.2    | 835.2  | 837.4  |
| F12_S05 | central_dpp     | mid  | P3   | 30    | 1.0000 | 1.0000 | 82.5     | 150.1  | 155.5  |
| F12_S05 | central_dpp     | high | P1   | 90    | 0.9111 | 0.0000 | 1946.2   | 1994.5 | 1999.5 |
| F12_S05 | central_dpp     | high | P2   | 90    | 0.9778 | 0.0000 | 792.1    | 861.7  | 865.5  |
| F12_S05 | central_dpp     | high | P3   | 90    | 1.0000 | 1.0000 | 77.4     | 141.8  | 146.7  |
| F12_S05 | shielded_dpp    | low  | P1   | 0     | –      | –      | 1893.0   | 1978.6 | 1982.3 |
| F12_S05 | shielded_dpp    | low  | P2   | 0     | –      | –      | 673.0    | 754.2  | 757.7  |
| F12_S05 | shielded_dpp    | low  | P3   | 0     | –      | –      | 84.9     | 152.9  | 159.8  |
| F12_S05 | shielded_dpp    | mid  | P1   | 30    | 0.5667 | 0.0000 | 1903.8   | 1973.6 | 1976.1 |
| F12_S05 | shielded_dpp    | mid  | P2   | 30    | 0.8333 | 0.0000 | 822.2    | 890.1  | 892.5  |
| F12_S05 | shielded_dpp    | mid  | P3   | 30    | 1.0000 | 1.0000 | 81.1     | 146.8  | 152.3  |
| F12_S05 | shielded_dpp    | high | P1   | 90    | 0.9222 | 0.0000 | 1914.0   | 1990.0 | 1992.6 |
| F12_S05 | shielded_dpp    | high | P2   | 90    | 0.9667 | 0.0000 | 701.7    | 774.2  | 778.5  |
| F12_S05 | shielded_dpp    | high | P3   | 90    | 1.0000 | 1.0000 | 80.0     | 140.8  | 147.1  |
| F12_S06 | fixed_local     | low  | P1   | 0     | –      | –      | 1932.5   | 2004.1 | 2017.5 |
| F12_S06 | fixed_local     | low  | P2   | 0     | –      | –      | 752.8    | 824.8  | 837.3  |
| F12_S06 | fixed_local     | low  | P3   | 0     | –      | –      | 90.0     | 164.6  | 171.9  |
| F12_S06 | fixed_local     | mid  | P1   | 0     | –      | –      | 1937.2   | 2042.4 | 2059.8 |
| F12_S06 | fixed_local     | mid  | P2   | 0     | –      | –      | 673.6    | 769.8  | 782.5  |
| F12_S06 | fixed_local     | mid  | P3   | 0     | –      | –      | 90.2     | 159.5  | 166.9  |
| F12_S06 | fixed_local     | high | P1   | 0     | –      | –      | 1909.3   | 1956.4 | 1959.2 |
| F12_S06 | fixed_local     | high | P2   | 0     | –      | –      | 785.0    | 826.4  | 829.8  |
| F12_S06 | fixed_local     | high | P3   | 0     | –      | –      | 90.5     | 162.6  | 169.1  |
| F12_S06 | fixed_summary   | low  | P1   | 0     | –      | –      | 1785.0   | 1821.6 | 1825.8 |
| F12_S06 | fixed_summary   | low  | P2   | 0     | –      | –      | 614.8    | 652.1  | 654.9  |
| F12_S06 | fixed_summary   | low  | P3   | 0     | –      | –      | 88.8     | 157.6  | 164.1  |
| F12_S06 | fixed_summary   | mid  | P1   | 0     | –      | –      | 1889.4   | 1925.6 | 1929.1 |
| F12_S06 | fixed_summary   | mid  | P2   | 0     | –      | –      | 741.5    | 779.2  | 781.7  |
| F12_S06 | fixed_summary   | mid  | P3   | 0     | –      | –      | 85.5     | 156.6  | 163.6  |
| F12_S06 | fixed_summary   | high | P1   | 0     | –      | –      | 1854.5   | 1988.9 | 2023.9 |
| F12_S06 | fixed_summary   | high | P2   | 0     | –      | –      | 659.0    | 780.9  | 817.9  |
| F12_S06 | fixed_summary   | high | P3   | 0     | –      | –      | 91.7     | 165.0  | 171.2  |
| F12_S06 | fixed_full      | low  | P1   | 90    | 0.0000 | 0.0000 | 2001.6   | 2059.9 | 2106.9 |
| F12_S06 | fixed_full      | low  | P2   | 90    | 0.0000 | 0.0000 | 694.5    | 809.4  | 850.0  |
| F12_S06 | fixed_full      | low  | P3   | 90    | 0.0000 | 0.0000 | 80.2     | 143.3  | 148.5  |
| F12_S06 | fixed_full      | mid  | P1   | 90    | 0.0000 | 0.0000 | 1890.5   | 2019.8 | 2023.1 |
| F12_S06 | fixed_full      | mid  | P2   | 90    | 0.0000 | 0.0000 | 652.4    | 715.4  | 735.8  |
| F12_S06 | fixed_full      | mid  | P3   | 90    | 0.0000 | 0.0000 | 77.1     | 141.6  | 146.7  |
| F12_S06 | fixed_full      | high | P1   | 90    | 0.9667 | 0.0000 | 1966.0   | 2015.7 | 2038.6 |
| F12_S06 | fixed_full      | high | P2   | 90    | 0.9111 | 0.0000 | 734.7    | 797.3  | 799.7  |
| F12_S06 | fixed_full      | high | P3   | 90    | 1.0000 | 1.0000 | 80.4     | 144.3  | 150.8  |
| F12_S06 | independent_dpp | low  | P1   | 22    | 0.0000 | 0.0000 | 2005.8   | 2040.9 | 2043.7 |
| F12_S06 | independent_dpp | low  | P2   | 22    | 0.0000 | 0.0000 | 696.2    | 797.4  | 800.0  |
| F12_S06 | independent_dpp | low  | P3   | 22    | 0.0000 | 0.0000 | 84.9     | 156.4  | 161.7  |
| F12_S06 | independent_dpp | mid  | P1   | 45    | 0.0000 | 0.0000 | 1969.6   | 1996.1 | 2000.6 |
| F12_S06 | independent_dpp | mid  | P2   | 45    | 0.0000 | 0.0000 | 903.6    | 932.0  | 934.8  |
| F12_S06 | independent_dpp | mid  | P3   | 45    | 0.0000 | 0.0000 | 79.1     | 144.5  | 150.1  |
| F12_S06 | independent_dpp | high | P1   | 90    | 0.9222 | 0.0000 | 1912.8   | 2042.7 | 2046.9 |
| F12_S06 | independent_dpp | high | P2   | 90    | 0.9444 | 0.0000 | 730.8    | 856.4  | 859.5  |
| F12_S06 | independent_dpp | high | P3   | 90    | 1.0000 | 1.0000 | 73.9     | 132.8  | 138.6  |
| F12_S06 | central_dpp     | low  | P1   | 0     | –      | –      | 1945.7   | 2067.9 | 2070.5 |
| F12_S06 | central_dpp     | low  | P2   | 0     | –      | –      | 746.1    | 825.0  | 827.8  |
| F12_S06 | central_dpp     | low  | P3   | 0     | –      | –      | 84.7     | 151.8  | 158.0  |
| F12_S06 | central_dpp     | mid  | P1   | 30    | 0.9000 | 0.0000 | 2001.6   | 2059.9 | 2070.6 |
| F12_S06 | central_dpp     | mid  | P2   | 30    | 0.8667 | 0.0000 | 869.7    | 935.4  | 937.7  |
| F12_S06 | central_dpp     | mid  | P3   | 30    | 1.0000 | 1.0000 | 89.2     | 151.7  | 158.0  |
| F12_S06 | central_dpp     | high | P1   | 90    | 0.9444 | 0.0000 | 2175.3   | 2214.4 | 2217.3 |
| F12_S06 | central_dpp     | high | P2   | 90    | 0.9667 | 0.0000 | 1059.9   | 1102.3 | 1105.7 |
| F12_S06 | central_dpp     | high | P3   | 90    | 1.0000 | 1.0000 | 79.4     | 154.0  | 160.2  |
| F12_S06 | shielded_dpp    | low  | P1   | 0     | –      | –      | 1933.6   | 1972.0 | 1976.6 |
| F12_S06 | shielded_dpp    | low  | P2   | 0     | –      | –      | 775.1    | 817.3  | 820.9  |
| F12_S06 | shielded_dpp    | low  | P3   | 0     | –      | –      | 89.9     | 160.3  | 166.5  |
| F12_S06 | shielded_dpp    | mid  | P1   | 30    | 0.8333 | 0.0000 | 1830.6   | 1960.8 | 1963.5 |
| F12_S06 | shielded_dpp    | mid  | P2   | 30    | 0.9000 | 0.0000 | 688.1    | 761.8  | 764.0  |
| F12_S06 | shielded_dpp    | mid  | P3   | 30    | 1.0000 | 1.0000 | 82.1     | 150.9  | 156.6  |
| F12_S06 | shielded_dpp    | high | P1   | 90    | 0.9778 | 0.0000 | 1922.9   | 1970.4 | 1972.8 |
| F12_S06 | shielded_dpp    | high | P2   | 90    | 0.9556 | 0.0000 | 698.8    | 740.1  | 744.4  |

| Session | Method          | Cap. | Dev. | $n_c$ | Post   | Nom.   | Late p50 | p95    | p99    |
|---------|-----------------|------|------|-------|--------|--------|----------|--------|--------|
| F12_S06 | shielded_dpp    | high | P3   | 90    | 1.0000 | 1.0000 | 88.2     | 149.6  | 155.1  |
| F12_S07 | fixed_local     | low  | P1   | 0     | –      | –      | 1944.9   | 2011.2 | 2014.7 |
| F12_S07 | fixed_local     | low  | P2   | 0     | –      | –      | 730.1    | 809.6  | 812.8  |
| F12_S07 | fixed_local     | low  | P3   | 0     | –      | –      | 88.8     | 160.6  | 167.3  |
| F12_S07 | fixed_local     | mid  | P1   | 0     | –      | –      | 1868.9   | 1943.8 | 1948.2 |
| F12_S07 | fixed_local     | mid  | P2   | 0     | –      | –      | 629.0    | 699.1  | 702.5  |
| F12_S07 | fixed_local     | mid  | P3   | 0     | –      | –      | 93.9     | 164.7  | 171.4  |
| F12_S07 | fixed_local     | high | P1   | 0     | –      | –      | 2100.5   | 2245.8 | 2249.1 |
| F12_S07 | fixed_local     | high | P2   | 0     | –      | –      | 895.1    | 1020.1 | 1024.0 |
| F12_S07 | fixed_local     | high | P3   | 0     | –      | –      | 88.5     | 159.7  | 166.7  |
| F12_S07 | fixed_summary   | low  | P1   | 0     | –      | –      | 1997.9   | 2132.3 | 2136.3 |
| F12_S07 | fixed_summary   | low  | P2   | 0     | –      | –      | 675.1    | 801.3  | 804.5  |
| F12_S07 | fixed_summary   | low  | P3   | 0     | –      | –      | 84.1     | 151.1  | 156.9  |
| F12_S07 | fixed_summary   | mid  | P1   | 0     | –      | –      | 1927.0   | 1978.8 | 1985.2 |
| F12_S07 | fixed_summary   | mid  | P2   | 0     | –      | –      | 612.5    | 680.9  | 684.0  |
| F12_S07 | fixed_summary   | mid  | P3   | 0     | –      | –      | 87.5     | 157.8  | 165.0  |
| F12_S07 | fixed_summary   | high | P1   | 0     | –      | –      | 2181.8   | 2223.8 | 2227.5 |
| F12_S07 | fixed_summary   | high | P2   | 0     | –      | –      | 613.1    | 751.0  | 755.5  |
| F12_S07 | fixed_summary   | high | P3   | 0     | –      | –      | 84.2     | 154.1  | 161.1  |
| F12_S07 | fixed_full      | low  | P1   | 90    | 0.0000 | 0.0000 | 1825.8   | 1945.2 | 1947.3 |
| F12_S07 | fixed_full      | low  | P2   | 90    | 0.0000 | 0.0000 | 557.2    | 701.7  | 705.8  |
| F12_S07 | fixed_full      | low  | P3   | 90    | 0.0000 | 0.0000 | 81.1     | 143.6  | 149.2  |
| F12_S07 | fixed_full      | mid  | P1   | 90    | 0.0000 | 0.0000 | 2023.6   | 2274.3 | 2277.8 |
| F12_S07 | fixed_full      | mid  | P2   | 90    | 0.0000 | 0.0000 | 771.6    | 952.8  | 955.2  |
| F12_S07 | fixed_full      | mid  | P3   | 90    | 0.0000 | 0.0000 | 77.4     | 139.8  | 145.4  |
| F12_S07 | fixed_full      | high | P1   | 90    | 0.9778 | 0.0000 | 1983.4   | 2025.4 | 2028.5 |
| F12_S07 | fixed_full      | high | P2   | 90    | 0.9333 | 0.0000 | 793.0    | 850.9  | 872.5  |
| F12_S07 | fixed_full      | high | P3   | 90    | 1.0000 | 1.0000 | 79.3     | 145.9  | 150.9  |
| F12_S07 | independent_dpp | low  | P1   | 22    | 0.0000 | 0.0000 | 1909.6   | 2101.8 | 2103.8 |
| F12_S07 | independent_dpp | low  | P2   | 22    | 0.0000 | 0.0000 | 717.8    | 874.3  | 876.8  |
| F12_S07 | independent_dpp | low  | P3   | 22    | 0.0000 | 0.0000 | 83.9     | 149.3  | 155.4  |
| F12_S07 | independent_dpp | mid  | P1   | 45    | 0.0000 | 0.0000 | 2049.3   | 2104.3 | 2107.5 |
| F12_S07 | independent_dpp | mid  | P2   | 45    | 0.0000 | 0.0000 | 941.2    | 966.4  | 970.3  |
| F12_S07 | independent_dpp | mid  | P3   | 45    | 0.0000 | 0.0000 | 85.9     | 151.4  | 156.4  |
| F12_S07 | independent_dpp | high | P1   | 90    | 0.9667 | 0.0000 | 1967.2   | 2170.5 | 2176.7 |
| F12_S07 | independent_dpp | high | P2   | 90    | 0.8889 | 0.0000 | 782.6    | 931.8  | 1045.7 |
| F12_S07 | independent_dpp | high | P3   | 90    | 0.9889 | 0.9889 | 81.2     | 328.2  | 333.1  |
| F12_S07 | central_dpp     | low  | P1   | 0     | –      | –      | 2036.6   | 2077.1 | 2082.3 |
| F12_S07 | central_dpp     | low  | P2   | 0     | –      | –      | 723.5    | 764.1  | 768.0  |
| F12_S07 | central_dpp     | low  | P3   | 0     | –      | –      | 88.3     | 156.0  | 161.8  |
| F12_S07 | central_dpp     | mid  | P1   | 30    | 0.8667 | 0.0000 | 2137.4   | 2178.3 | 2188.0 |
| F12_S07 | central_dpp     | mid  | P2   | 30    | 0.8667 | 0.0000 | 861.1    | 927.6  | 931.2  |
| F12_S07 | central_dpp     | mid  | P3   | 30    | 1.0000 | 1.0000 | 87.2     | 159.0  | 164.3  |
| F12_S07 | central_dpp     | high | P1   | 90    | 0.9778 | 0.0000 | 2061.6   | 2089.0 | 2093.5 |
| F12_S07 | central_dpp     | high | P2   | 90    | 0.9333 | 0.0000 | 864.5    | 941.0  | 944.8  |
| F12_S07 | central_dpp     | high | P3   | 90    | 1.0000 | 1.0000 | 81.4     | 144.8  | 150.1  |
| F12_S07 | shielded_dpp    | low  | P1   | 0     | –      | –      | 1894.6   | 1993.4 | 1996.3 |
| F12_S07 | shielded_dpp    | low  | P2   | 0     | –      | –      | 628.4    | 734.7  | 738.1  |
| F12_S07 | shielded_dpp    | low  | P3   | 0     | –      | –      | 85.6     | 158.3  | 163.7  |
| F12_S07 | shielded_dpp    | mid  | P1   | 30    | 0.9333 | 0.0000 | 2068.2   | 2134.6 | 2137.8 |
| F12_S07 | shielded_dpp    | mid  | P2   | 30    | 0.9333 | 0.0000 | 716.9    | 752.4  | 755.3  |
| F12_S07 | shielded_dpp    | mid  | P3   | 30    | 1.0000 | 1.0000 | 79.0     | 149.3  | 154.3  |
| F12_S07 | shielded_dpp    | high | P1   | 90    | 0.9889 | 0.0000 | 1891.0   | 1956.7 | 1959.9 |
| F12_S07 | shielded_dpp    | high | P2   | 90    | 0.9556 | 0.0000 | 704.9    | 781.2  | 833.6  |
| F12_S07 | shielded_dpp    | high | P3   | 90    | 1.0000 | 1.0000 | 79.4     | 144.9  | 151.5  |
| F12_S08 | fixed_local     | low  | P1   | 0     | –      | –      | 1957.4   | 2013.6 | 2016.1 |
| F12_S08 | fixed_local     | low  | P2   | 0     | –      | –      | 685.9    | 732.8  | 735.9  |
| F12_S08 | fixed_local     | low  | P3   | 0     | –      | –      | 87.0     | 158.9  | 165.4  |
| F12_S08 | fixed_local     | mid  | P1   | 0     | –      | –      | 1817.2   | 1937.7 | 1954.1 |
| F12_S08 | fixed_local     | mid  | P2   | 0     | –      | –      | 614.7    | 724.4  | 744.3  |
| F12_S08 | fixed_local     | mid  | P3   | 0     | –      | –      | 88.0     | 158.5  | 164.9  |
| F12_S08 | fixed_local     | high | P1   | 0     | –      | –      | 1937.6   | 2017.9 | 2035.3 |
| F12_S08 | fixed_local     | high | P2   | 0     | –      | –      | 683.8    | 776.3  | 784.7  |
| F12_S08 | fixed_local     | high | P3   | 0     | –      | –      | 89.9     | 164.2  | 170.0  |
| F12_S08 | fixed_summary   | low  | P1   | 0     | –      | –      | 1999.1   | 2079.8 | 2083.7 |
| F12_S08 | fixed_summary   | low  | P2   | 0     | –      | –      | 735.8    | 778.9  | 786.1  |
| F12_S08 | fixed_summary   | low  | P3   | 0     | –      | –      | 88.0     | 158.4  | 164.5  |
| F12_S08 | fixed_summary   | mid  | P1   | 0     | –      | –      | 1992.5   | 2077.9 | 2082.7 |
| F12_S08 | fixed_summary   | mid  | P2   | 0     | –      | –      | 745.7    | 848.4  | 854.3  |
| F12_S08 | fixed_summary   | mid  | P3   | 0     | –      | –      | 89.3     | 158.2  | 164.7  |
| F12_S08 | fixed_summary   | high | P1   | 0     | –      | –      | 2038.2   | 2116.8 | 2119.5 |
| F12_S08 | fixed_summary   | high | P2   | 0     | –      | –      | 750.9    | 807.1  | 811.8  |
| F12_S08 | fixed_summary   | high | P3   | 0     | –      | –      | 87.3     | 161.2  | 167.8  |
| F12_S08 | fixed_full      | low  | P1   | 90    | 0.0000 | 0.0000 | 2158.3   | 2203.4 | 2206.0 |
| F12_S08 | fixed_full      | low  | P2   | 90    | 0.0000 | 0.0000 | 978.1    | 1030.7 | 1034.2 |
| F12_S08 | fixed_full      | low  | P3   | 90    | 0.0000 | 0.0000 | 75.1     | 136.5  | 142.3  |
| F12_S08 | fixed_full      | mid  | P1   | 90    | 0.0000 | 0.0000 | 1956.9   | 2041.8 | 2044.2 |
| F12_S08 | fixed_full      | mid  | P2   | 90    | 0.0000 | 0.0000 | 904.9    | 1014.7 | 1018.5 |
| F12_S08 | fixed_full      | mid  | P3   | 90    | 0.0000 | 0.0000 | 80.4     | 142.8  | 147.6  |
| F12_S08 | fixed_full      | high | P1   | 90    | 0.9333 | 0.0000 | 2053.0   | 2190.7 | 2193.1 |
| F12_S08 | fixed_full      | high | P2   | 90    | 0.9556 | 0.0000 | 908.0    | 1033.1 | 1036.6 |
| F12_S08 | fixed_full      | high | P3   | 90    | 1.0000 | 1.0000 | 80.5     | 142.0  | 146.8  |
| F12_S08 | independent_dpp | low  | P1   | 22    | 0.0000 | 0.0000 | 1874.7   | 2010.2 | 2012.1 |

| Session | Method          | Cap. | Dev. | $n_c$ | Post   | Nom.   | Late p50 | p95    | p99    |
|---------|-----------------|------|------|-------|--------|--------|----------|--------|--------|
| F12_S08 | independent_dpp | low  | P2   | 22    | 0.0000 | 0.0000 | 773.6    | 889.6  | 892.9  |
| F12_S08 | independent_dpp | low  | P3   | 22    | 0.0000 | 0.0000 | 82.8     | 150.6  | 158.3  |
| F12_S08 | independent_dpp | mid  | P1   | 45    | 0.0000 | 0.0000 | 1854.8   | 1911.5 | 1913.6 |
| F12_S08 | independent_dpp | mid  | P2   | 45    | 0.0000 | 0.0000 | 664.3    | 721.6  | 724.3  |
| F12_S08 | independent_dpp | mid  | P3   | 45    | 0.0000 | 0.0000 | 78.2     | 140.8  | 145.7  |
| F12_S08 | independent_dpp | high | P1   | 90    | 0.9222 | 0.0000 | 2146.9   | 2229.6 | 2231.4 |
| F12_S08 | independent_dpp | high | P2   | 90    | 0.9333 | 0.0000 | 995.3    | 1104.8 | 1107.4 |
| F12_S08 | independent_dpp | high | P3   | 90    | 1.0000 | 1.0000 | 75.3     | 141.2  | 146.0  |
| F12_S08 | central_dpp     | low  | P1   | 0     | –      | –      | 2023.5   | 2362.2 | 2365.6 |
| F12_S08 | central_dpp     | low  | P2   | 0     | –      | –      | 750.6    | 1083.7 | 1087.1 |
| F12_S08 | central_dpp     | low  | P3   | 0     | –      | –      | 83.0     | 154.3  | 161.6  |
| F12_S08 | central_dpp     | mid  | P1   | 30    | 0.9667 | 0.0000 | 2057.4   | 2105.0 | 2107.2 |
| F12_S08 | central_dpp     | mid  | P2   | 30    | 0.6667 | 0.0000 | 784.2    | 856.1  | 858.2  |
| F12_S08 | central_dpp     | mid  | P3   | 30    | 1.0000 | 1.0000 | 79.3     | 146.0  | 152.4  |
| F12_S08 | central_dpp     | high | P1   | 90    | 0.9222 | 0.0000 | 2048.7   | 2112.0 | 2115.0 |
| F12_S08 | central_dpp     | high | P2   | 90    | 0.9556 | 0.0000 | 814.7    | 873.6  | 877.0  |
| F12_S08 | central_dpp     | high | P3   | 90    | 1.0000 | 1.0000 | 80.4     | 143.1  | 148.8  |
| F12_S08 | shielded_dpp    | low  | P1   | 0     | –      | –      | 1979.4   | 2072.9 | 2075.3 |
| F12_S08 | shielded_dpp    | low  | P2   | 0     | –      | –      | 746.4    | 815.6  | 819.0  |
| F12_S08 | shielded_dpp    | low  | P3   | 0     | –      | –      | 87.2     | 154.6  | 160.7  |
| F12_S08 | shielded_dpp    | mid  | P1   | 30    | 0.7333 | 0.0000 | 1895.4   | 1977.8 | 1980.2 |
| F12_S08 | shielded_dpp    | mid  | P2   | 30    | 0.7000 | 0.0000 | 754.9    | 834.4  | 837.4  |
| F12_S08 | shielded_dpp    | mid  | P3   | 30    | 1.0000 | 1.0000 | 84.4     | 149.1  | 155.6  |
| F12_S08 | shielded_dpp    | high | P1   | 90    | 0.9556 | 0.0000 | 2070.2   | 2145.7 | 2150.4 |
| F12_S08 | shielded_dpp    | high | P2   | 90    | 0.9556 | 0.0000 | 946.3    | 1008.9 | 1011.7 |
| F12_S08 | shielded_dpp    | high | P3   | 90    | 1.0000 | 1.0000 | 72.2     | 135.7  | 140.7  |
| F12_S09 | fixed_local     | low  | P1   | 0     | –      | –      | 2062.2   | 2148.9 | 2153.1 |
| F12_S09 | fixed_local     | low  | P2   | 0     | –      | –      | 835.3    | 908.6  | 912.9  |
| F12_S09 | fixed_local     | low  | P3   | 0     | –      | –      | 91.8     | 166.5  | 174.5  |
| F12_S09 | fixed_local     | mid  | P1   | 0     | –      | –      | 2017.6   | 2068.9 | 2073.1 |
| F12_S09 | fixed_local     | mid  | P2   | 0     | –      | –      | 734.4    | 788.0  | 793.2  |
| F12_S09 | fixed_local     | mid  | P3   | 0     | –      | –      | 84.3     | 154.2  | 160.5  |
| F12_S09 | fixed_local     | high | P1   | 0     | –      | –      | 2105.4   | 2162.6 | 2166.2 |
| F12_S09 | fixed_local     | high | P2   | 0     | –      | –      | 862.7    | 933.5  | 937.1  |
| F12_S09 | fixed_local     | high | P3   | 0     | –      | –      | 90.6     | 162.1  | 167.7  |
| F12_S09 | fixed_summary   | low  | P1   | 0     | –      | –      | 2035.6   | 2133.6 | 2137.1 |
| F12_S09 | fixed_summary   | low  | P2   | 0     | –      | –      | 740.9    | 828.5  | 832.8  |
| F12_S09 | fixed_summary   | low  | P3   | 0     | –      | –      | 91.4     | 166.4  | 172.2  |
| F12_S09 | fixed_summary   | mid  | P1   | 0     | –      | –      | 2067.9   | 2207.0 | 2209.7 |
| F12_S09 | fixed_summary   | mid  | P2   | 0     | –      | –      | 731.5    | 865.4  | 869.0  |
| F12_S09 | fixed_summary   | mid  | P3   | 0     | –      | –      | 89.9     | 163.7  | 169.8  |
| F12_S09 | fixed_summary   | high | P1   | 0     | –      | –      | 1931.7   | 2076.8 | 2079.4 |
| F12_S09 | fixed_summary   | high | P2   | 0     | –      | –      | 656.3    | 775.4  | 778.8  |
| F12_S09 | fixed_summary   | high | P3   | 0     | –      | –      | 84.7     | 152.8  | 158.7  |
| F12_S09 | fixed_full      | low  | P1   | 90    | 0.0000 | 0.0000 | 2058.8   | 2117.3 | 2124.4 |
| F12_S09 | fixed_full      | low  | P2   | 90    | 0.0000 | 0.0000 | 756.7    | 835.5  | 843.0  |
| F12_S09 | fixed_full      | low  | P3   | 90    | 0.0000 | 0.0000 | 77.2     | 141.2  | 148.3  |
| F12_S09 | fixed_full      | mid  | P1   | 90    | 0.0000 | 0.0000 | 2005.3   | 2062.5 | 2089.6 |
| F12_S09 | fixed_full      | mid  | P2   | 90    | 0.0000 | 0.0000 | 877.8    | 931.8  | 935.3  |
| F12_S09 | fixed_full      | mid  | P3   | 90    | 0.0000 | 0.0000 | 77.9     | 136.7  | 142.7  |
| F12_S09 | fixed_full      | high | P1   | 90    | 0.9111 | 0.0000 | 2041.2   | 2101.3 | 2106.6 |
| F12_S09 | fixed_full      | high | P2   | 90    | 0.9556 | 0.0000 | 795.5    | 870.5  | 874.5  |
| F12_S09 | fixed_full      | high | P3   | 90    | 1.0000 | 1.0000 | 77.1     | 141.8  | 147.0  |
| F12_S09 | independent_dpp | low  | P1   | 22    | 0.0000 | 0.0000 | 2044.8   | 2140.4 | 2142.8 |
| F12_S09 | independent_dpp | low  | P2   | 22    | 0.0000 | 0.0000 | 886.4    | 937.7  | 959.9  |
| F12_S09 | independent_dpp | low  | P3   | 22    | 0.0000 | 0.0000 | 85.2     | 148.8  | 157.8  |
| F12_S09 | independent_dpp | mid  | P1   | 45    | 0.0000 | 0.0000 | 2163.0   | 2195.9 | 2197.6 |
| F12_S09 | independent_dpp | mid  | P2   | 45    | 0.0000 | 0.0000 | 1015.9   | 1045.1 | 1065.2 |
| F12_S09 | independent_dpp | mid  | P3   | 45    | 0.0000 | 0.0000 | 78.9     | 140.5  | 145.3  |
| F12_S09 | independent_dpp | high | P1   | 90    | 0.9333 | 0.0000 | 2125.5   | 2172.7 | 2174.9 |
| F12_S09 | independent_dpp | high | P2   | 90    | 0.9333 | 0.0000 | 851.2    | 955.2  | 959.5  |
| F12_S09 | independent_dpp | high | P3   | 90    | 1.0000 | 1.0000 | 74.2     | 135.0  | 139.5  |
| F12_S09 | central_dpp     | low  | P1   | 0     | –      | –      | 1974.6   | 2062.7 | 2073.2 |
| F12_S09 | central_dpp     | low  | P2   | 0     | –      | –      | 700.8    | 778.4  | 783.7  |
| F12_S09 | central_dpp     | low  | P3   | 0     | –      | –      | 88.6     | 160.8  | 166.0  |
| F12_S09 | central_dpp     | mid  | P1   | 30    | 0.9000 | 0.0000 | 2105.6   | 2140.7 | 2143.5 |
| F12_S09 | central_dpp     | mid  | P2   | 30    | 0.7667 | 0.0000 | 957.1    | 996.2  | 1018.1 |
| F12_S09 | central_dpp     | mid  | P3   | 30    | 1.0000 | 1.0000 | 85.9     | 153.4  | 158.5  |
| F12_S09 | central_dpp     | high | P1   | 90    | 0.9556 | 0.0000 | 2328.7   | 2368.8 | 2373.6 |
| F12_S09 | central_dpp     | high | P2   | 90    | 0.9778 | 0.0000 | 997.1    | 1036.6 | 1039.8 |
| F12_S09 | central_dpp     | high | P3   | 90    | 1.0000 | 1.0000 | 68.1     | 135.1  | 142.1  |
| F12_S09 | shielded_dpp    | low  | P1   | 0     | –      | –      | 1968.5   | 2040.6 | 2046.2 |
| F12_S09 | shielded_dpp    | low  | P2   | 0     | –      | –      | 790.4    | 868.6  | 870.5  |
| F12_S09 | shielded_dpp    | low  | P3   | 0     | –      | –      | 92.7     | 162.5  | 168.8  |
| F12_S09 | shielded_dpp    | mid  | P1   | 30    | 0.8667 | 0.0000 | 2063.5   | 2130.1 | 2140.9 |
| F12_S09 | shielded_dpp    | mid  | P2   | 30    | 0.9000 | 0.0000 | 834.2    | 885.0  | 888.3  |
| F12_S09 | shielded_dpp    | mid  | P3   | 30    | 1.0000 | 1.0000 | 80.4     | 145.5  | 150.6  |
| F12_S09 | shielded_dpp    | high | P1   | 90    | 0.9444 | 0.0000 | 2087.2   | 2138.8 | 2140.9 |
| F12_S09 | shielded_dpp    | high | P2   | 90    | 0.9444 | 0.0000 | 824.1    | 897.0  | 899.7  |
| F12_S09 | shielded_dpp    | high | P3   | 90    | 1.0000 | 1.0000 | 73.9     | 137.8  | 143.0  |
| F12_S10 | fixed_local     | low  | P1   | 0     | –      | –      | 1963.5   | 2064.8 | 2068.2 |
| F12_S10 | fixed_local     | low  | P2   | 0     | –      | –      | 761.3    | 837.0  | 839.8  |
| F12_S10 | fixed_local     | low  | P3   | 0     | –      | –      | 91.5     | 166.4  | 173.2  |

| Session | Method          | Cap. | Dev. | $n_c$ | Post   | Nom.   | Late p50 | p95    | p99    |
|---------|-----------------|------|------|-------|--------|--------|----------|--------|--------|
| F12_S10 | fixed_local     | mid  | P1   | 0     | –      | –      | 2011.0   | 2050.3 | 2053.7 |
| F12_S10 | fixed_local     | mid  | P2   | 0     | –      | –      | 715.1    | 755.4  | 758.7  |
| F12_S10 | fixed_local     | mid  | P3   | 0     | –      | –      | 90.9     | 162.7  | 169.3  |
| F12_S10 | fixed_local     | high | P1   | 0     | –      | –      | 1937.4   | 2000.6 | 2003.6 |
| F12_S10 | fixed_local     | high | P2   | 0     | –      | –      | 668.6    | 723.4  | 726.7  |
| F12_S10 | fixed_local     | high | P3   | 0     | –      | –      | 93.9     | 163.6  | 170.2  |
| F12_S10 | fixed_summary   | low  | P1   | 0     | –      | –      | 2013.5   | 2103.8 | 2108.3 |
| F12_S10 | fixed_summary   | low  | P2   | 0     | –      | –      | 823.8    | 913.5  | 917.0  |
| F12_S10 | fixed_summary   | low  | P3   | 0     | –      | –      | 84.6     | 153.6  | 160.5  |
| F12_S10 | fixed_summary   | mid  | P1   | 0     | –      | –      | 2057.6   | 2136.9 | 2140.5 |
| F12_S10 | fixed_summary   | mid  | P2   | 0     | –      | –      | 946.8    | 1014.3 | 1018.1 |
| F12_S10 | fixed_summary   | mid  | P3   | 0     | –      | –      | 84.4     | 155.9  | 163.6  |
| F12_S10 | fixed_summary   | high | P1   | 0     | –      | –      | 1943.4   | 2129.7 | 2134.3 |
| F12_S10 | fixed_summary   | high | P2   | 0     | –      | –      | 764.8    | 939.7  | 943.1  |
| F12_S10 | fixed_summary   | high | P3   | 0     | –      | –      | 87.3     | 163.1  | 170.0  |
| F12_S10 | fixed_full      | low  | P1   | 90    | 0.0000 | 0.0000 | 1967.7   | 2037.9 | 2041.3 |
| F12_S10 | fixed_full      | low  | P2   | 90    | 0.0000 | 0.0000 | 858.5    | 951.5  | 953.8  |
| F12_S10 | fixed_full      | low  | P3   | 90    | 0.0000 | 0.0000 | 76.8     | 138.7  | 143.3  |
| F12_S10 | fixed_full      | mid  | P1   | 90    | 0.0000 | 0.0000 | 2146.3   | 2221.4 | 2223.5 |
| F12_S10 | fixed_full      | mid  | P2   | 90    | 0.0000 | 0.0000 | 847.3    | 890.1  | 894.4  |
| F12_S10 | fixed_full      | mid  | P3   | 90    | 0.0000 | 0.0000 | 74.8     | 132.3  | 136.9  |
| F12_S10 | fixed_full      | high | P1   | 90    | 0.9778 | 0.0000 | 2165.6   | 2199.9 | 2204.4 |
| F12_S10 | fixed_full      | high | P2   | 90    | 0.9667 | 0.0000 | 871.9    | 912.1  | 916.5  |
| F12_S10 | fixed_full      | high | P3   | 90    | 1.0000 | 1.0000 | 81.3     | 142.9  | 147.6  |
| F12_S10 | independent_dpp | low  | P1   | 22    | 0.0000 | 0.0000 | 1921.7   | 2011.9 | 2016.7 |
| F12_S10 | independent_dpp | low  | P2   | 22    | 0.0000 | 0.0000 | 653.8    | 788.2  | 791.3  |
| F12_S10 | independent_dpp | low  | P3   | 22    | 0.0000 | 0.0000 | 82.9     | 150.7  | 156.6  |
| F12_S10 | independent_dpp | mid  | P1   | 45    | 0.0000 | 0.0000 | 1887.2   | 1966.4 | 1968.3 |
| F12_S10 | independent_dpp | mid  | P2   | 45    | 0.0000 | 0.0000 | 773.0    | 838.0  | 840.0  |
| F12_S10 | independent_dpp | mid  | P3   | 45    | 0.0000 | 0.0000 | 75.8     | 141.7  | 147.3  |
| F12_S10 | independent_dpp | high | P1   | 90    | 0.8556 | 0.0000 | 1965.1   | 2090.0 | 2093.2 |
| F12_S10 | independent_dpp | high | P2   | 90    | 0.9667 | 0.0000 | 882.8    | 981.7  | 984.1  |
| F12_S10 | independent_dpp | high | P3   | 90    | 1.0000 | 1.0000 | 76.0     | 142.0  | 147.0  |
| F12_S10 | central_dpp     | low  | P1   | 0     | –      | –      | 2041.7   | 2115.9 | 2119.5 |
| F12_S10 | central_dpp     | low  | P2   | 0     | –      | –      | 834.2    | 905.3  | 909.7  |
| F12_S10 | central_dpp     | low  | P3   | 0     | –      | –      | 89.3     | 164.3  | 169.0  |
| F12_S10 | central_dpp     | mid  | P1   | 30    | 0.9667 | 0.0000 | 2076.9   | 2105.3 | 2151.2 |
| F12_S10 | central_dpp     | mid  | P2   | 30    | 0.9000 | 0.0000 | 850.4    | 929.6  | 931.8  |
| F12_S10 | central_dpp     | mid  | P3   | 30    | 1.0000 | 1.0000 | 80.3     | 148.5  | 154.8  |
| F12_S10 | central_dpp     | high | P1   | 90    | 0.9556 | 0.0000 | 2040.7   | 2102.1 | 2106.1 |
| F12_S10 | central_dpp     | high | P2   | 90    | 0.9667 | 0.0000 | 905.9    | 940.3  | 944.7  |
| F12_S10 | central_dpp     | high | P3   | 90    | 1.0000 | 1.0000 | 76.4     | 140.4  | 146.7  |
| F12_S10 | shielded_dpp    | low  | P1   | 0     | –      | –      | 2085.7   | 2128.5 | 2130.9 |
| F12_S10 | shielded_dpp    | low  | P2   | 0     | –      | –      | 938.2    | 977.5  | 981.4  |
| F12_S10 | shielded_dpp    | low  | P3   | 0     | –      | –      | 89.8     | 160.9  | 166.9  |
| F12_S10 | shielded_dpp    | mid  | P1   | 30    | 0.8000 | 0.0000 | 1980.5   | 2053.7 | 2058.1 |
| F12_S10 | shielded_dpp    | mid  | P2   | 30    | 0.7667 | 0.0000 | 729.0    | 827.8  | 830.2  |
| F12_S10 | shielded_dpp    | mid  | P3   | 30    | 1.0000 | 1.0000 | 86.9     | 154.8  | 160.6  |
| F12_S10 | shielded_dpp    | high | P1   | 90    | 0.9222 | 0.0000 | 2020.4   | 2096.1 | 2123.6 |
| F12_S10 | shielded_dpp    | high | P2   | 90    | 0.9778 | 0.0000 | 879.9    | 928.6  | 932.5  |
| F12_S10 | shielded_dpp    | high | P3   | 90    | 1.0000 | 1.0000 | 80.1     | 143.5  | 149.0  |
| F12_S11 | fixed_local     | low  | P1   | 0     | –      | –      | 1923.5   | 1986.9 | 2001.2 |
| F12_S11 | fixed_local     | low  | P2   | 0     | –      | –      | 743.9    | 820.1  | 835.6  |
| F12_S11 | fixed_local     | low  | P3   | 0     | –      | –      | 89.1     | 161.4  | 168.1  |
| F12_S11 | fixed_local     | mid  | P1   | 0     | –      | –      | 2032.4   | 2109.7 | 2112.1 |
| F12_S11 | fixed_local     | mid  | P2   | 0     | –      | –      | 748.3    | 824.6  | 828.8  |
| F12_S11 | fixed_local     | mid  | P3   | 0     | –      | –      | 93.9     | 169.5  | 176.2  |
| F12_S11 | fixed_local     | high | P1   | 0     | –      | –      | 2031.0   | 2105.3 | 2109.9 |
| F12_S11 | fixed_local     | high | P2   | 0     | –      | –      | 736.1    | 814.3  | 818.7  |
| F12_S11 | fixed_local     | high | P3   | 0     | –      | –      | 90.1     | 162.4  | 168.3  |
| F12_S11 | fixed_summary   | low  | P1   | 0     | –      | –      | 1928.4   | 1983.9 | 1990.1 |
| F12_S11 | fixed_summary   | low  | P2   | 0     | –      | –      | 639.4    | 714.9  | 725.7  |
| F12_S11 | fixed_summary   | low  | P3   | 0     | –      | –      | 82.4     | 153.2  | 159.2  |
| F12_S11 | fixed_summary   | mid  | P1   | 0     | –      | –      | 2087.9   | 2136.3 | 2140.6 |
| F12_S11 | fixed_summary   | mid  | P2   | 0     | –      | –      | 824.7    | 887.7  | 894.9  |
| F12_S11 | fixed_summary   | mid  | P3   | 0     | –      | –      | 86.8     | 157.3  | 162.5  |
| F12_S11 | fixed_summary   | high | P1   | 0     | –      | –      | 1941.6   | 1992.7 | 1995.2 |
| F12_S11 | fixed_summary   | high | P2   | 0     | –      | –      | 648.9    | 703.8  | 707.1  |
| F12_S11 | fixed_summary   | high | P3   | 0     | –      | –      | 86.9     | 155.8  | 163.5  |
| F12_S11 | fixed_full      | low  | P1   | 90    | 0.0000 | 0.0000 | 1942.0   | 2002.2 | 2006.4 |
| F12_S11 | fixed_full      | low  | P2   | 90    | 0.0000 | 0.0000 | 724.1    | 806.1  | 808.5  |
| F12_S11 | fixed_full      | low  | P3   | 90    | 0.0000 | 0.0000 | 85.2     | 150.6  | 155.4  |
| F12_S11 | fixed_full      | mid  | P1   | 90    | 0.0000 | 0.0000 | 2070.3   | 2149.2 | 2152.6 |
| F12_S11 | fixed_full      | mid  | P2   | 90    | 0.0000 | 0.0000 | 938.8    | 980.4  | 983.0  |
| F12_S11 | fixed_full      | mid  | P3   | 90    | 0.0000 | 0.0000 | 79.7     | 142.6  | 148.2  |
| F12_S11 | fixed_full      | high | P1   | 90    | 0.9111 | 0.0000 | 2113.8   | 2201.4 | 2203.3 |
| F12_S11 | fixed_full      | high | P2   | 90    | 0.9333 | 0.0000 | 912.5    | 1038.8 | 1042.4 |
| F12_S11 | fixed_full      | high | P3   | 90    | 1.0000 | 1.0000 | 76.8     | 135.8  | 141.6  |
| F12_S11 | independent_dpp | low  | P1   | 22    | 0.0000 | 0.0000 | 1986.8   | 2019.7 | 2022.3 |
| F12_S11 | independent_dpp | low  | P2   | 22    | 0.0000 | 0.0000 | 764.0    | 802.3  | 804.8  |
| F12_S11 | independent_dpp | low  | P3   | 22    | 0.0000 | 0.0000 | 88.9     | 156.0  | 162.7  |
| F12_S11 | independent_dpp | mid  | P1   | 45    | 0.0000 | 0.0000 | 2001.6   | 2087.2 | 2091.6 |
| F12_S11 | independent_dpp | mid  | P2   | 45    | 0.0000 | 0.0000 | 824.6    | 904.2  | 906.9  |

| Session | Method          | Cap. | Dev. | $n_c$ | Post   | Nom.   | Late p50 | p95    | p99    |
|---------|-----------------|------|------|-------|--------|--------|----------|--------|--------|
| F12_S11 | independent_dpp | mid  | P3   | 45    | 0.0000 | 0.0000 | 78.9     | 141.5  | 146.5  |
| F12_S11 | independent_dpp | high | P1   | 90    | 0.9000 | 0.0000 | 2133.6   | 2182.2 | 2184.9 |
| F12_S11 | independent_dpp | high | P2   | 90    | 0.9444 | 0.0000 | 981.2    | 1046.9 | 1051.6 |
| F12_S11 | independent_dpp | high | P3   | 90    | 1.0000 | 1.0000 | 71.8     | 129.6  | 136.5  |
| F12_S11 | central_dpp     | low  | P1   | 0     | –      | –      | 2043.7   | 2116.0 | 2120.4 |
| F12_S11 | central_dpp     | low  | P2   | 0     | –      | –      | 783.2    | 865.3  | 868.8  |
| F12_S11 | central_dpp     | low  | P3   | 0     | –      | –      | 93.5     | 162.4  | 168.8  |
| F12_S11 | central_dpp     | mid  | P1   | 30    | 0.8000 | 0.0000 | 2106.5   | 2167.1 | 2169.4 |
| F12_S11 | central_dpp     | mid  | P2   | 30    | 0.8333 | 0.0000 | 934.4    | 991.7  | 995.4  |
| F12_S11 | central_dpp     | mid  | P3   | 30    | 1.0000 | 1.0000 | 80.0     | 146.7  | 151.1  |
| F12_S11 | central_dpp     | high | P1   | 90    | 0.8889 | 0.0000 | 1923.1   | 1979.1 | 1982.6 |
| F12_S11 | central_dpp     | high | P2   | 90    | 0.9444 | 0.0000 | 689.4    | 782.6  | 785.7  |
| F12_S11 | central_dpp     | high | P3   | 90    | 1.0000 | 1.0000 | 75.7     | 135.1  | 140.8  |
| F12_S11 | shielded_dpp    | low  | P1   | 0     | –      | –      | 2014.5   | 2077.7 | 2080.4 |
| F12_S11 | shielded_dpp    | low  | P2   | 0     | –      | –      | 735.4    | 818.5  | 822.9  |
| F12_S11 | shielded_dpp    | low  | P3   | 0     | –      | –      | 87.8     | 158.6  | 165.2  |
| F12_S11 | shielded_dpp    | mid  | P1   | 30    | 0.8333 | 0.0000 | 1884.2   | 1947.6 | 1950.1 |
| F12_S11 | shielded_dpp    | mid  | P2   | 30    | 0.8667 | 0.0000 | 802.2    | 834.0  | 882.2  |
| F12_S11 | shielded_dpp    | mid  | P3   | 30    | 1.0000 | 1.0000 | 79.7     | 142.8  | 149.4  |
| F12_S11 | shielded_dpp    | high | P1   | 90    | 0.9000 | 0.0000 | 1866.9   | 1926.8 | 1928.9 |
| F12_S11 | shielded_dpp    | high | P2   | 90    | 0.9333 | 0.0000 | 651.9    | 721.0  | 724.2  |
| F12_S11 | shielded_dpp    | high | P3   | 90    | 1.0000 | 1.0000 | 78.9     | 143.3  | 149.7  |
| F12_S12 | fixed_local     | low  | P1   | 0     | –      | –      | 2021.6   | 2060.8 | 2064.9 |
| F12_S12 | fixed_local     | low  | P2   | 0     | –      | –      | 851.3    | 891.4  | 896.0  |
| F12_S12 | fixed_local     | low  | P3   | 0     | –      | –      | 89.4     | 162.6  | 169.8  |
| F12_S12 | fixed_local     | mid  | P1   | 0     | –      | –      | 2065.4   | 2110.6 | 2113.4 |
| F12_S12 | fixed_local     | mid  | P2   | 0     | –      | –      | 880.8    | 918.2  | 921.2  |
| F12_S12 | fixed_local     | mid  | P3   | 0     | –      | –      | 95.8     | 167.4  | 173.3  |
| F12_S12 | fixed_local     | high | P1   | 0     | –      | –      | 2028.6   | 2158.6 | 2162.3 |
| F12_S12 | fixed_local     | high | P2   | 0     | –      | –      | 775.1    | 837.5  | 843.8  |
| F12_S12 | fixed_local     | high | P3   | 0     | –      | –      | 92.3     | 164.3  | 171.5  |
| F12_S12 | fixed_summary   | low  | P1   | 0     | –      | –      | 1910.3   | 1985.2 | 1989.2 |
| F12_S12 | fixed_summary   | low  | P2   | 0     | –      | –      | 718.9    | 806.6  | 808.8  |
| F12_S12 | fixed_summary   | low  | P3   | 0     | –      | –      | 88.1     | 159.9  | 165.1  |
| F12_S12 | fixed_summary   | mid  | P1   | 0     | –      | –      | 1893.9   | 1959.5 | 1962.4 |
| F12_S12 | fixed_summary   | mid  | P2   | 0     | –      | –      | 613.9    | 675.0  | 678.6  |
| F12_S12 | fixed_summary   | mid  | P3   | 0     | –      | –      | 88.1     | 155.1  | 161.6  |
| F12_S12 | fixed_summary   | high | P1   | 0     | –      | –      | 2217.8   | 2256.9 | 2260.5 |
| F12_S12 | fixed_summary   | high | P2   | 0     | –      | –      | 1057.1   | 1098.5 | 1102.7 |
| F12_S12 | fixed_summary   | high | P3   | 0     | –      | –      | 82.4     | 150.6  | 156.6  |
| F12_S12 | fixed_full      | low  | P1   | 90    | 0.0000 | 0.0000 | 2056.0   | 2127.3 | 2128.7 |
| F12_S12 | fixed_full      | low  | P2   | 90    | 0.0000 | 0.0000 | 831.5    | 884.5  | 889.1  |
| F12_S12 | fixed_full      | low  | P3   | 90    | 0.0000 | 0.0000 | 85.1     | 157.7  | 163.5  |
| F12_S12 | fixed_full      | mid  | P1   | 90    | 0.0000 | 0.0000 | 1859.9   | 1997.8 | 2002.3 |
| F12_S12 | fixed_full      | mid  | P2   | 90    | 0.0000 | 0.0000 | 697.3    | 841.4  | 844.7  |
| F12_S12 | fixed_full      | mid  | P3   | 90    | 0.0000 | 0.0000 | 83.0     | 145.3  | 151.3  |
| F12_S12 | fixed_full      | high | P1   | 90    | 0.9222 | 0.0000 | 2070.3   | 2135.0 | 2139.0 |
| F12_S12 | fixed_full      | high | P2   | 90    | 0.9444 | 0.0000 | 956.1    | 1001.6 | 1003.5 |
| F12_S12 | fixed_full      | high | P3   | 90    | 1.0000 | 1.0000 | 80.0     | 138.6  | 145.2  |
| F12_S12 | independent_dpp | low  | P1   | 22    | 0.0000 | 0.0000 | 1001.7   | 1038.7 | 1041.5 |
| F12_S12 | independent_dpp | low  | P2   | 22    | 0.0000 | 0.0000 | 864.1    | 915.7  | 918.1  |
| F12_S12 | independent_dpp | low  | P3   | 22    | 0.0000 | 0.0000 | 85.0     | 157.5  | 162.6  |
| F12_S12 | independent_dpp | mid  | P1   | 45    | 0.0000 | 0.0000 | 2161.5   | 2193.0 | 2197.7 |
| F12_S12 | independent_dpp | mid  | P2   | 45    | 0.0000 | 0.0000 | 860.3    | 896.0  | 897.8  |
| F12_S12 | independent_dpp | mid  | P3   | 45    | 0.0000 | 0.0000 | 77.0     | 138.3  | 143.6  |
| F12_S12 | independent_dpp | high | P1   | 90    | 0.9444 | 0.0000 | 2004.9   | 2052.8 | 2056.1 |
| F12_S12 | independent_dpp | high | P2   | 90    | 0.9444 | 0.0000 | 778.7    | 819.1  | 822.4  |
| F12_S12 | independent_dpp | high | P3   | 90    | 1.0000 | 1.0000 | 74.8     | 135.8  | 141.5  |
| F12_S12 | central_dpp     | low  | P1   | 0     | –      | –      | 2072.6   | 2369.1 | 2372.7 |
| F12_S12 | central_dpp     | low  | P2   | 0     | –      | –      | 865.3    | 1186.7 | 1191.2 |
| F12_S12 | central_dpp     | low  | P3   | 0     | –      | –      | 86.1     | 161.1  | 167.6  |
| F12_S12 | central_dpp     | mid  | P1   | 30    | 0.5333 | 0.0000 | 1883.9   | 1935.5 | 1939.8 |
| F12_S12 | central_dpp     | mid  | P2   | 30    | 0.7667 | 0.0000 | 659.2    | 732.8  | 735.7  |
| F12_S12 | central_dpp     | mid  | P3   | 30    | 1.0000 | 1.0000 | 83.3     | 144.6  | 150.7  |
| F12_S12 | central_dpp     | high | P1   | 90    | 0.9222 | 0.0000 | 2095.8   | 2187.1 | 2190.5 |
| F12_S12 | central_dpp     | high | P2   | 90    | 0.9444 | 0.0000 | 846.9    | 919.8  | 922.5  |
| F12_S12 | central_dpp     | high | P3   | 90    | 1.0000 | 1.0000 | 81.8     | 147.6  | 152.4  |
| F12_S12 | shielded_dpp    | low  | P1   | 0     | –      | –      | 1879.5   | 1960.7 | 1964.6 |
| F12_S12 | shielded_dpp    | low  | P2   | 0     | –      | –      | 695.3    | 756.2  | 759.7  |
| F12_S12 | shielded_dpp    | low  | P3   | 0     | –      | –      | 87.3     | 156.0  | 161.3  |
| F12_S12 | shielded_dpp    | mid  | P1   | 30    | 0.9000 | 0.0000 | 2041.9   | 2073.2 | 2075.5 |
| F12_S12 | shielded_dpp    | mid  | P2   | 30    | 0.9333 | 0.0000 | 803.7    | 835.2  | 837.5  |
| F12_S12 | shielded_dpp    | mid  | P3   | 30    | 1.0000 | 1.0000 | 83.1     | 147.1  | 152.7  |
| F12_S12 | shielded_dpp    | high | P1   | 90    | 0.9444 | 0.0000 | 1962.8   | 2023.5 | 2026.7 |
| F12_S12 | shielded_dpp    | high | P2   | 90    | 0.9444 | 0.0000 | 736.5    | 774.2  | 776.1  |
| F12_S12 | shielded_dpp    | high | P3   | 90    | 1.0000 | 1.0000 | 74.7     | 136.9  | 142.8  |

### VIII. RELEASE-LATENESS DIAGNOSTIC

Independent nominal clocks, the three-probe barrier, unequal first-iteration waiting, relative sleep, and no absolute catch-up are source/log supported. The specific OS/network reason for the recurring proposal-arrival order is UNKNOWN.

TABLE VI: Release-latency diagnostic from accepted formal timestamps (ms).

| Device | Events | p50     | p95     | p99     | Unproven residual                                                                                                                             |
|--------|--------|---------|---------|---------|-----------------------------------------------------------------------------------------------------------------------------------------------|
| P1     | 19440  | 1972.34 | 2173.36 | 2326.13 | The exact OS/network reason that proposals repeatedly reached the barrier in P3/P2/P1 order, and residual per-session variation, are UNKNOWN. |
| P2     | 19440  | 763.90  | 976.78  | 1080.30 | The exact OS/network reason that proposals repeatedly reached the barrier in P3/P2/P1 order, and residual per-session variation, are UNKNOWN. |
| P3     | 19440  | 83.56   | 154.08  | 167.94  | The exact OS/network reason that proposals repeatedly reached the barrier in P3/P2/P1 order, and residual per-session variation, are UNKNOWN. |

# IX. FINITE- $n$ SCALING CONFORMANCE

This is discrete-event conformance, not a 32-probe hardware scalability experiment.

TABLE VII: All finite- $n$  discrete-event conformance cells.

| $n$ | $\rho$ | $C$ (B) | $k^*$ | Sim. $k$ | Theory frac. | Sim. frac. | Match |
|-----|--------|---------|-------|----------|--------------|------------|-------|
| 3   | 1/3    | 131072  | 0     | 0        | 0.000000     | 0.000000   | PASS  |
| 3   | 1/2    | 196608  | 1     | 1        | 0.333333     | 0.333333   | PASS  |
| 3   | 2/3    | 262144  | 1     | 1        | 0.333333     | 0.333333   | PASS  |
| 3   | 3/4    | 294912  | 2     | 2        | 0.666667     | 0.666667   | PASS  |
| 3   | 1/1    | 393216  | 3     | 3        | 1.000000     | 1.000000   | PASS  |
| 8   | 1/3    | 349525  | 1     | 1        | 0.125000     | 0.125000   | PASS  |
| 8   | 1/2    | 524288  | 3     | 3        | 0.375000     | 0.375000   | PASS  |
| 8   | 2/3    | 699050  | 4     | 4        | 0.500000     | 0.500000   | PASS  |
| 8   | 3/4    | 786432  | 5     | 5        | 0.625000     | 0.625000   | PASS  |
| 8   | 1/1    | 1048576 | 8     | 8        | 1.000000     | 1.000000   | PASS  |
| 16  | 1/3    | 699050  | 3     | 3        | 0.187500     | 0.187500   | PASS  |
| 16  | 1/2    | 1048576 | 6     | 6        | 0.375000     | 0.375000   | PASS  |
| 16  | 2/3    | 1398101 | 9     | 9        | 0.562500     | 0.562500   | PASS  |
| 16  | 3/4    | 1572864 | 11    | 11       | 0.687500     | 0.687500   | PASS  |
| 16  | 1/1    | 2097152 | 16    | 16       | 1.000000     | 1.000000   | PASS  |
| 32  | 1/3    | 1398101 | 7     | 7        | 0.218750     | 0.218750   | PASS  |
| 32  | 1/2    | 2097152 | 13    | 13       | 0.406250     | 0.406250   | PASS  |
| 32  | 2/3    | 2796202 | 19    | 19       | 0.593750     | 0.593750   | PASS  |
| 32  | 3/4    | 3145728 | 22    | 22       | 0.687500     | 0.687500   | PASS  |
| 32  | 1/1    | 4194304 | 32    | 32       | 1.000000     | 1.000000   | PASS  |

## X. COMMON-DENOMINATOR CONTROLLER COMPARISON

The main paper’s mid-capacity comparison divides on-time full-core results by all 270 generated measured events. It is derived directly from the frozen phase-sensitivity CSV rather than from conditional coverage alone. The coordinated policy

TABLE VIII  
MID-CAPACITY COMMON-DENOMINATOR SOFTWARE SENSITIVITY.

| Controller    | Full req. | Timely/gen. | Cond. | MB     |
|---------------|-----------|-------------|-------|--------|
| C-QPG         | 0.333     | 0.3333      | 1.000 | 14.746 |
| Desync. I-QPG | 0.667     | 0.3352      | 0.503 | 23.593 |
| Sync. I-QPG   | 0.500     | 0.0000      | 0.000 | 17.695 |

uses 37.5% fewer application bytes than the phase- randomized independent policy (14.7456 versus 23.59296 MB) while producing 0.3333 versus 0.3352 timely full results per generated event.

## XI. INDEPENDENCE APPROXIMATION

At mid capacity, an independent Bernoulli request model with  $p = 2/3$  gives  $P(J = 1) = 6/27$ ,  $P(J = 2) = 12/27$ , and  $P(J = 3) = 8/27$ . If a  $2B$  service budget finishes all requests for  $J \leq 2$  and none for  $J = 3$ , the expected conditional timely fraction is  $5/9 \approx 0.5556$  and the common timely-full fraction is  $10/27 \approx 0.3704$ . The phase-randomized simulator gives 0.5028 and 0.3352, respectively. The Bernoulli calculation is an independence approximation, not a proved upper bound.

## XII. ROTATING-FRONTIER SOFTWARE CONFORMANCE

The cyclic construction uses the same 1-ms integer service accounting as the phase study. It admits exactly  $K = \lfloor C/B \rfloor$  full requests per round and zero payload from the other probes. There are zero simulated shared-cap violations. This discrete-event

TABLE IX  
IDEALIZED ROTATING-FRONTIER CONFORMANCE.

| Capacity | $K$ | Full fraction | Timely full / generated | Fair period |
|----------|-----|---------------|-------------------------|-------------|
| Low      | 1   | 0.3333        | 0.3333                  | 3           |
| Mid      | 2   | 0.6667        | 0.6667                  | 3           |
| High     | 3   | 1.0000        | 1.0000                  | 1           |

check was completed before the separately frozen Raspberry Pi extension and is retained only as a conformance check on the cyclic byte-allocation rule; the physical results are reported separately below.

## XIII. CRFA HARDWARE FRONTIER EXTENSION

The separately frozen CRFA campaign tests the same cyclic construction on the three physical probes without changing the primary A–F campaign. Twelve matched session/workload identities are executed at each capacity, yielding 36 accepted arms. Across 55 indexed attempts, 19 attempts are excluded for predefined infrastructure failures; all excluded attempts are retained in the attempt index, timing outcomes are never acceptance criteria, and partial attempts are not stitched into accepted arms. The accepted extension contains 9,720 measured occurrences and 10,620 reconstructed shared-service rounds. All frozen event-identity, action/payload, final-drain, raw-reconstruction, and shared-cap gates pass, with zero shared-cap violations.

TABLE X  
CRFA PHYSICAL ATTAINMENT OF THE UNRESTRICTED FULL-EVIDENCE BYTE FRONTIER. THE C-QPG COLUMN USES THE SESSION-MATCHED PRIMARY HARDWARE RESULT; PAIRED INTERVALS ARE DESCRIPTIVE 95% SESSION-BOOTSTRAP INTERVALS.

| Capacity | Byte frontier | C-QPG  | CRFA   | Attainment | $\Delta$ CRFA–C-QPG (95% CI) |
|----------|---------------|--------|--------|------------|------------------------------|
| Low      | 0.3333        | 0.0000 | 0.3056 | 0.9167     | +0.3056 [+0.3015, +0.3093]   |
| Mid      | 0.6667        | 0.2938 | 0.6327 | 0.9491     | +0.3389 [+0.3306, +0.3478]   |
| High     | 1.0000        | 0.9698 | 0.9660 | 0.9660     | -0.0037 [-0.0133, +0.0046]   |

At low, mid, and high capacity, CRFA produces 0.3056, 0.6327, and 0.9660 timely full-core results per generated event, corresponding to 91.7%, 94.9%, and 96.6% of the byte frontier. Every selected full-evidence result is eventually returned; the residual to the byte frontier is therefore post-release timing loss rather than loss of selected evidence. At mid capacity, CRFA produces 2,050 timely full-core results from 3,240 generated occurrences, compared with 952 for session-matched C-QPG. The paired mean difference is +0.3389 with 95% session-bootstrap interval [0.3306, 0.3478]. At high capacity, where universal coverage no longer costs a full-evidence slot, CRFA–C-QPG is  $-0.0037$  with interval  $[-0.0133, 0.0046]$ .

These results are not a universal dominance claim. CRFA assigns H-NONE to non-selected probes and therefore exchanges same-round remote coverage for more full-evidence opportunities. C-QPG instead preserves at least summary evidence for every probe in every round.

#### XIV. CRFA PER-DEVICE ADMISSION FAIRNESS

The cyclic construction equalizes full-evidence opportunity counts exactly, while the physical post-release timing of those opportunities remains device dependent. Across the 12 accepted sessions, each probe receives 360, 720, and 1,080 full-evidence

TABLE XI  
CRFA PER-DEVICE ADMISSION OPPORTUNITY AND POST-RELEASE TIMING.

| Capacity | Probe | Full opps. | Timely full | Cond. timely | Max sel. interval |
|----------|-------|------------|-------------|--------------|-------------------|
| Low      | P1    | 360        | 314         | 0.8722       | 3                 |
| Low      | P2    | 360        | 316         | 0.8778       | 3                 |
| Low      | P3    | 360        | 360         | 1.0000       | 3                 |
| Mid      | P1    | 720        | 659         | 0.9153       | 2                 |
| Mid      | P2    | 720        | 671         | 0.9319       | 2                 |
| Mid      | P3    | 720        | 720         | 1.0000       | 2                 |
| High     | P1    | 1080       | 1021        | 0.9454       | 1                 |
| High     | P2    | 1080       | 1030        | 0.9537       | 1                 |
| High     | P3    | 1080       | 1079        | 0.9991       | 1                 |

opportunities at low, mid, and high capacity, respectively. The maximum inter-selection interval is three, two, and one measured rounds. At mid capacity, conditional post-release on-time fractions for selected full events are 0.9153, 0.9319, and 1.0000 for P1, P2, and P3. Thus deterministic admission fairness does not imply identical physical timing.

#### XV. CRFA PER-DEVICE TIMING RECONSTRUCTION

TABLE XII  
CRFA SELECTED-FULL TIMING BY DEVICE. NOMINAL ON-TIME REFERS TO THE SCHEDULE-INCLUSIVE ANCHOR; POST-RELEASE ON-TIME USES ACTUAL EVENT CREATION.

| Capacity | Probe | Post timely | Nom. timely | Release p50 (ms) | Post p50 (ms) | Post p95 (ms) |
|----------|-------|-------------|-------------|------------------|---------------|---------------|
| Low      | P1    | 0.8722      | 0.0000      | 2182.8           | 938.1         | 1042.3        |
| Low      | P2    | 0.8778      | 0.0000      | 1077.9           | 925.9         | 1028.7        |
| Low      | P3    | 1.0000      | 1.0000      | 82.2             | 245.4         | 340.6         |
| Mid      | P1    | 0.9153      | 0.0000      | 2142.1           | 928.2         | 1011.2        |
| Mid      | P2    | 0.9319      | 0.0000      | 903.5            | 886.0         | 1012.5        |
| Mid      | P3    | 1.0000      | 1.0000      | 78.4             | 238.4         | 338.9         |
| High     | P1    | 0.9454      | 0.0000      | 5079.6           | 900.2         | 1000.9        |
| High     | P2    | 0.9537      | 0.0000      | 3621.6           | 879.9         | 998.5         |
| High     | P3    | 0.9991      | 0.9991      | 77.0             | 238.8         | 333.0         |

P1 and P2 have zero nominal-anchor timely full results at all three capacities in the CRFA extension, whereas P3 is essentially fully nominally on time. The aggregate CRFA nominal timely-full fractions on the common generated-event denominator are 0.1111, 0.2222, and 0.3330 at low, mid, and high capacity, versus post-release values of 0.3056, 0.6327, and 0.9660. This reproduces the schedule-anchor asymmetry seen in the primary campaign and reinforces the scope of the primary 1-s quantity as a post-release service SLO.

#### XVI. CRFA PER-SESSION RESULTS

TABLE XIII  
CRFA SESSION-LEVEL TIMELY FULL-CORE OUTPUT AND FRONTIER ATTAINMENT.

| Session | Low timely | Low $\eta$ | Mid timely | Mid $\eta$ | High timely | High $\eta$ |
|---------|------------|------------|------------|------------|-------------|-------------|
| F12_S01 | 0.3000     | 0.9000     | 0.6259     | 0.9389     | 0.9704      | 0.9704      |
| F12_S02 | 0.3000     | 0.9000     | 0.6333     | 0.9500     | 0.9519      | 0.9519      |
| F12_S03 | 0.3074     | 0.9222     | 0.6111     | 0.9167     | 0.9481      | 0.9481      |
| F12_S04 | 0.3074     | 0.9222     | 0.6296     | 0.9444     | 0.9704      | 0.9704      |
| F12_S05 | 0.3111     | 0.9333     | 0.6259     | 0.9389     | 0.9667      | 0.9667      |
| F12_S06 | 0.2926     | 0.8778     | 0.6333     | 0.9500     | 0.9852      | 0.9852      |
| F12_S07 | 0.3037     | 0.9111     | 0.6370     | 0.9556     | 0.9667      | 0.9667      |
| F12_S08 | 0.3074     | 0.9222     | 0.6407     | 0.9611     | 0.9519      | 0.9519      |
| F12_S09 | 0.3111     | 0.9333     | 0.6370     | 0.9556     | 0.9815      | 0.9815      |
| F12_S10 | 0.3111     | 0.9333     | 0.6519     | 0.9778     | 0.9778      | 0.9778      |
| F12_S11 | 0.3185     | 0.9556     | 0.6407     | 0.9611     | 0.9667      | 0.9667      |
| F12_S12 | 0.2963     | 0.8889     | 0.6259     | 0.9389     | 0.9556      | 0.9556      |

## XVII. HARDWARE TIMING ANCHOR SUMMARY

TABLE XIV  
ACCEPTED HARDWARE CORE-REQUIRED TIMING BY PROBE.

| Probe | Post  | Nominal | Lateness p50 (ms) |
|-------|-------|---------|-------------------|
| P1    | 0.583 | 0.000   | 1972.3            |
| P2    | 0.588 | 0.000   | 763.9             |
| P3    | 0.630 | 0.630   | 83.6              |

P1 and P2 have zero nominal-anchor on-time coverage when all accepted core-required occurrences are pooled by device. P1’s median release offset is about 1.97 s and therefore exceeds the 1-s service horizon before post-release service time is added. The exact OS/network cause of the recurring proposal arrival order remains unknown.

## XVIII. ACCEPTED AND EXCLUDED ATTEMPTS

For the primary A–F campaign, accepted attempts are S01-a01, S02-a01, S03-a02, S04-a02, S05-a01, S06-a01, S07-a02, and S08–S12-a01. S03-a01, S04-a01, and S07-a01 are excluded infrastructure attempts and are not stitched into accepted sessions.

The CRFA extension indexes 55 attempts for 36 planned arms. Thirty-six arms are accepted and 19 attempts are excluded under the frozen infrastructure rules, predominantly SSH transport failures plus a pre-arm hardware-gate failure. The accepted and excluded lineage is preserved in the extension attempt index. Timing or favorable scientific outcomes are not retry criteria.

## XIX. DETECTOR AND REPLAY LIMITATIONS

The workload is frozen CICIOT2023-derived feature-vector replay, not packet replay or an uncontrolled deployment. The dataset is cited in the main paper. The formal occurrence set excludes Recon-PortScan, and exact feature-vector overlap exists across frozen partitions. Perfect correctness on the formal occurrences is therefore a controlled diagnostic rather than an IDS-superiority result.

## XX. RAW AUDIT AND OFFLINE-ANALYSIS VERDICTS

The primary full-raw audit rebuilt 216/216 accepted arms and 58,320 measured events with zero raw-to-metric mismatch and zero shared-cap violation across 63,720 service rounds. Its historical verdict remains `PASS_WITH_ISSUES` because the raw bundle lacks direct artifacts for the S03-a01 SSD incident and the S07 pre-hardware launcher abort; the CRFA extension does not alter that disclosure.

The separately archived CRFA evidence reconstructs 36/36 accepted arms, 9,720 measured occurrences, and 10,620 shared-service rounds. Its frozen identity, action/payload, final-drain, reconstruction, and cap gates pass, and no shared-cap violation is observed. Across the two hardware campaigns this corresponds to 252 accepted arms, 68,040 measured occurrences, and 74,340 reconstructed shared-service rounds, with zero observed shared-cap violations.

The read-only offline analyses contain 432 queue-weight rows, 240 phase- sensitivity rows, 648 primary schedule-anchor rows, and 20 multi- $n$  scaling rows. The rotating software-conformance table remains a separate discrete- event check and is not relabelled as hardware evidence.

## XXI. ARTIFACT ACCESS

Machine-readable CSV/JSON sources, including the CRFA arm/capacity/session summaries, occurrence ledger, service audit, fairness table, timing reconstruction, matched comparisons, attempt index, number provenance, claim audit, build report, validation report, and SHA-256 manifest accompany this supplementary PDF. The CRFA primary archive and integrity-checked mirror reside on the same user-authorized external physical volume and therefore do not constitute independent physical backups. Third-party dataset redistribution rights remain outside the scope of this package.